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Theoretical Aspects of GPCR–Ligand Complex Pharmacology

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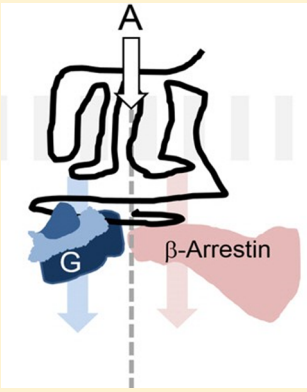
ABSTRACT: Over the past 50 years in pharmacology, an understanding of seven transmembrane (7TMR) function has been gained from the comparison of experimental data to receptor models. These models have been constructed from building blocks composed of systems consisting of series and parallel mass action binding reactions. Basic functions such as the the isomerization of receptors upon ligand binding, the sequential binding of receptors to membrane coupling proteins, and the selection of multiple receptor conformations have been combined in various ways to build receptor systems such as the ternary complex, extended ternary complex, and cubic ternary complex models for 7TMR function. Separately, the Black/Leff operational model has furnished an extremely valuable method of quantifying drug agonism. In the past few years, incorporation of the basic allosteric nature of 7TMRs has led to additional useful models of functional receptor allosteric mechanisms; these models yield valuable methods for quantifying allosteric effects. Finally, molecular dynamics has provided yet another new set of models describing the probability of formation of multiple receptor states; these radically new models are extremely useful in the prediction of functionally selective drug effects.



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"What is it that breathes fire into the equations and makes a universe for them to describe?..." —Stephen W. Hawking (1991)

1. INTRODUCTION: MODELS IN PHARMACOLOGY

The optimal approach for determining the mechanism of action (MOA) of any drug is to study the effects of an extensive range of concentrations of the drug in a functional system either directly or in a system probed by another drug such as an agonist. The resulting pattern of responses can then be compared to a quantitative mathematical model of drug–receptor interaction. If the data agree with the model, then the model is consistent with one possible MOA for the drug. Such agreement is not proof of the MOA; it indicates only consistency with one possible MOA. This is because, often, a number of mechanisms can provide similar patterns of concentration–response data. Progress in this process is made when the data do not agree with the predictions of the model. Under such circumstances, the model can be rejected or refined to give a new, and hopefully better, model. A number of types of models are available that can be classified in terms of complexity and ease of parameter estimation. Beginning with simplicity, the most simple models are *caricature models*, which contain obscure parameters and thus tend to have low estimability. More complex models become *heuristic models*, which capture the essential features of a system but lack the independent means to measure the required parameters. If estimable parameters are available, then *data-fitting models* utilizing mathematical equations can be used to fit experimental data. These types of models usually suffice to assess agreement toward a mechanism of action. However, the one-way aspect of experiments (consistent with but not exclusively definitive) of these types of approaches make it possible that an agreeable mechanism of action might still not be the correct one. Further on the complexity scale reside *simulation models*, which have a large number of parameters, not all of which can be determined

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from real data; these models are meant to reconstruct and represent reality.

The confidence that can be ascribed to any assessment of how appropriate a model is generally is made by determining how closely the predicted dose−response patterns fit the experimental data. This should be done with as large a range of concentrations of drugs as possible to detect a pattern. For example, a hallmark of an allosteric effect (whereby the ligand binds to a separate site on the receptor from the endogenous ligand) is saturation of the effect, that is, cessation of the effect when the allosteric binding site is completely occupied. Therefore, a wide range of concentrations of the test ligand should be tested to maximize the likelihood of detecting saturation of the effect and, thus, identifying an allosteric mechanism.

One of the main advantages inherent in the use of quantitative mathematical models is that they permit the prediction of behavior through rules that define the model. Adherence to all criteria predicted by a model must be absolute: If there is one exception, then the model is invalid. At this point, it is worth considering the models that have been proposed to describe receptor function in pharmacology. As a preface, it is useful to discuss the main building block for all of these models, namely, the mass action binding of a ligand to a protein.

2. MASS ACTION EQUATION

The root process for pharmacological analyses of drug−receptor interaction is found in the work of Guldberg and Waage,[1] who wrote

$$\mathrm{A} + \mathrm{B} \rightleftarrows \mathrm{A}' + \mathrm{B}' \tag{1}$$

This expression states that the rates of the reactions, both forward and backward, depend on the "active masses" (i.e., the concentrations) of reactants in the mixture ("...the amount of substance in the 'sphere of action' or, put another way, the concentration in the medium"[1]) . For these reasons, it is referred to as the "mass action" model. As applied to pharmacology, this scheme underpins all models of drug activity. Its main function is to relate the quantity of the drug ([A]) to the amount of the initiator of pharmacological activity in all physiological processes, namely, the drug−target complex. Assuming that drug A binds to receptor R with a rate of onset of k_1 (in s^{-1} M^{-1}), then the rate of drug−receptor association is given by $k_1[A][R]$. If the rate constant for dissociation of the drug from the receptor is given by k_2, then the rate of dissociation of the drug from the receptor is given by $k_2[AR]$. At equilibrium, the rate of association equals the rate of dissociation; the receptor conservation equation, namely, $[R] + [AR] = [R_T]$, where $[R_T]$ is the total amount of receptor, allows for the expression of [R]. Defining $K_A = k_2/k_1$ leads to

$$K_A = [A]([R_T] - [AR])/[AR] \tag{2}$$

This expression reduces to the pharmacological version of the mass action equation

$$[AR] = [A][R_T]/([A] + K_A) \tag{3}$$

Equation 3 describes the amount of product of the reaction (drug−target complex) as the ratio of the product of one of the reactants (drug concentration [A]) multiplied by the maximal output capability of the system (given by $[R_T]$) and the sum of the reactant ([A]) and a potency factor K_A. The value of K_A, defined as the equilibrium-dissociation constant for the drug−receptor complex, is a measure of affinity. Specifically, K_A is the concentration needed to bind to 50% of the available receptors; therefore, K_A^{-1} is the affinity. In terms of the pharmacological application of the mass action law, the following assumptions are made:

(1) The binding is reversible; violation of this assumption precludes calculation of valid K_A values
(2) All receptors are equally accessible to ligands; violation of this assumptions leads to incomplete assessment of ligand binding.
(3) Receptors are either free or bound to ligand, and there is not more than one affinity state or states of partial binding (the ligand and receptor must exist in only two states, bound or unbound).
(4) Binding does not alter the ligand or receptor; violation of this assumption also leads to ambiguity in the assignment of potency values (system-dependent potency). When a ligand binds to a receptor and changes its conformation, this is an expression of pharmacological efficacy.

The binding equation (eq 3) was derived by A. V. Hill.[2] Ten years later, eq 3 was again derived to describe the binding of molecules to a metal surface by Irving Langmuir, a chemist for General Electric who was interested in adsorbing chemicals to metal filaments for light bulbs.[3] Langmuir began with the assumption that molecules have an intrinsic property that will cause them to adsorb to a surface; he called this a process of "condensation", denoted by the symbol α. Therefore, in Langmuir's notation, the first-order rate of condensation is given by $\mu\alpha(1 - \theta_\mu)$, where μ is the concentration and θ_μ is the fraction of surface already bound by molecule. Langmuir also assumed that, once bound, molecules intrinsically diffused away from the surface, a process he called "evaporation", governed by a rate constant V_1; the rate of evaporation is $\theta_\mu V_1$. At equilibrium, one can write

$$\mu\alpha(1 - \theta_\mu) = \theta_\mu V_1 \tag{4}$$

Consolidation of terms leads to Langmuir's version of the adsorption isotherm

$$\theta_\mu = \alpha\mu/(\alpha\mu + V_1) \tag{5}$$

which corresponds to eq 3 with the substitutions $[AR]/[R_T] = \theta_\mu$ (fraction of receptor bound as a fraction of metal surface bound), $\mu = [A]$, and $K_A = V_1/\alpha$ [where $K_A = k_2/k_1$ is the rate of offset (evaporation) divided by the rate of onset (condensation)].

It will be seen that many pharmacological models do not adhere to the assumptions of the simple mass action binding equation, and thus, modifications of the original mass action model must be made (vide infra). In addition, a common observation in pharmacology, namely, the production of physiological response by ligands binding to receptors, required a major modification of standard binding models, namely, the introduction of ligand efficacy.

3. EFFICACY AND PHARMACOLOGICAL MODELS OF AGONISM

Some drugs bind to receptors and simply interfere with the function of hormones, neurotransmitters, and autacoids; these are termed antagonists. However, other drugs bind to receptors and induce a physiological response; these are termed agonists. Major departures from standard biochemical and enzymatic models of physiological function were required in pharmacol-

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ogy to accommodate the description of agonism and ligand efficacy. The major observation that had to be explained was the inability of simple binding equations to model physiological agonism. The binding equation (eq 3) predicts a sigmoidal curve on a semilogarithmic axis; this is shown by the dotted line in Figure 1. The horizontal axis in Figure 1 is the concentration

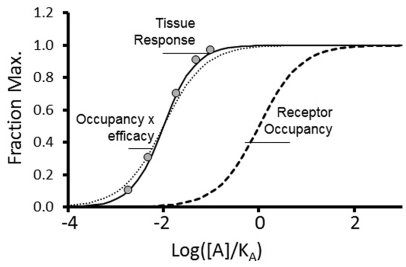

Figure 1. Relationship between curves describing the binding of a ligand to the receptor (through the adsorption isotherm, eq 3) and the observed functional response to an agonist (data from ref 6). The dotted line shows the binding occupancy curve multiplied by a constant factor of 100 to simulate that the ligand has an "efficacy" to scale each occupied receptor with 100 units of response. The curve modified by this efficacy factor closely simulates the observed response.

of ligand normalized by K_A; thus the binding curve will have a location parameter of 50% binding at $\log([A]/K_A) = 0$. However, experimentally, it was observed that some ligands produced a submaximal response (these are partial agonists), and thus, the ordinate values for the concentration−response curve did not reach 100% of the potential response scale of the assay. To accommodate partial agonists, Ariens and colleagues[4,5] introduced the parameter "intrinsic activity", denoted α. This is a multiplier of the binding curve equal to the observed maximal response of the agonist. Thus, if the agonist produced 50% of the tissue maximal response, α was set to 0.5. If an agonist produced the full maximal response capability of the system, $\alpha = 1$ and the ligand was considered a "full agonist". Although this device worked for partial agonists, it was inadequate for full agonists because another phenomenon was observed, namely, cases where the response curve is shifted to the left, along the concentration axis, of the receptor binding curve.

Figure 1 shows a pharmacological concentration−response curve to the agonist acetylcholine in guinea pig ileum.[6] It can be seen that the response curve is shifted to the left of the receptor binding curve by a factor of 100. This phenomenon is referred to as a "receptor reserve" in that it shows how the maximal response to such an agonist can be achieved through activation of only a fraction of the available receptors to produce the maximal system response (in Figure 1, the maximal response is achieved at a receptor occupancy of approximately 1%). These types of disparities clearly indicate that ligand binding is insufficient to describe pharmacological response. This led to modifications of the binding equation in pharmacology. Stephenson[7] introduced a multiplier that he termed "efficacy" for the ligand−receptor complex in the binding curve to account for receptor reserve; in essence, it allowed a single agonist−receptor complex to produce variable degrees of agonism in a system, where the magnitude of that power to induce a response was a result of the intrinsic power of the agonist to induce a response and also the sensitivity of the tissue. The amalgam of the ligand property and sensitivity of the system was referred to as efficacy[7] and denoted by e. Figure 1 shows a dotted line corresponding to a multiplier of 100 of the [AR] complex in the binding curve, which indicates an efficacy for acetylcholine in this system of 100. It can be seen that this multiplier models the observed tissue response. The introduction of Stephenson's efficacy was widely adopted in pharmacology as a useful tool for describing agonism, but it was an ad hoc solution to the problem of differences between binding and function and had little physiological basis. This shortcoming was alleviated by the introduction of the Black/Leff operational model for agonism.[8]

3.1. Black/Leff Operational Model

The model is based on the observation that agonist concentration−response curves are hyperbolic in nature. Therefore, the response can be expressed as

$$\text{response} = [A]E_{max}/([A] + \varphi) \quad (6)$$

where E_{max} is the maximal response of the system, [A] is the drug concentration, and φ is a location parameter for the curve along the concentration axis. An expression for [A] can then be derived as

$$[A] = \varphi\text{response}/(E_{max} - \text{response}) \quad (7)$$

An expression for [A] can also be derived from eq 3; relating this expression to eq 7 and rearranging gives

$$\text{response} = [A]E_{max}K_A/\{[A](K_A - \varphi) + [R_T]\varphi\} \quad (8)$$

Unacceptable negative or infinite values for responses would be observed for $K_A < \varphi$; therefore, either a linear relationship berween agonist concentration and response ($K_A = \varphi$) or a hyperbolic relationship ($K_A > \varphi$) is operative; this latter condition is most common in pharmacology. For instance, Figure 2 shows the hyperbolic relationship between the AR

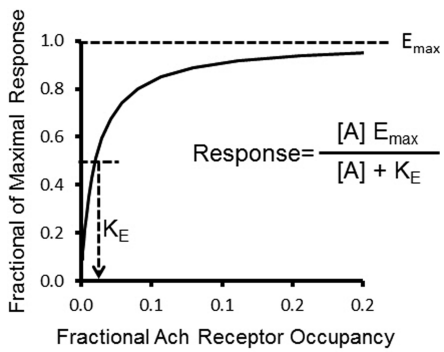

Figure 2. Hyperbolic relationship between the receptor occupancy of acetylcholine in guinea pig ileum and the tissue response (data from ref 6). This curve, determined from data, closely resembles one described by Michaelis−Menten kinetics for enzymes.

complex for acetylcholine and the muscarinic receptor in guinea pig ileum and the response from the curves shown in Figure 1. Therefore, assuming a hyperbolic realtionship beween receptor occupancy and response, the operational model gives the response as

$$\text{response} = [A][R_T]E_{max}/\{[A][R_T] + K_E + K_AK_E\} \quad (9)$$

where K_E is the equilibrium dissociation constant of the agonist−receptor complex and the response elements of the cell. A useful device is to define a term for efficacy as $[R_T]/K_E$, denoted by τ. This term represents the intrinsic power of the

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molecule to induce a response and also the efficiency of the system to transduce that stimulation into cell response. Thus, the equation for the operational model of agonism is[8]

$$\text{response} = [A]\tau E_{max}/\{[A](1 + \tau) + K_A\} \quad (10)$$

In essence, the operational model utilizes the cell as a virtual enzyme, with the AR complex functioning as the substrate and the tissue response functioning as the product. In these terms, the response can be modeled with the Michaelis−Menten equation[9,10]

$$\text{response} = [AR]E_{max}/([A] + K_E) \quad (11)$$

where the enzyme's maximum rate V_{max} is E_{max} and the Michaelis−Menten constant K_m is K_E. Black and Leff also published a version of the model that could handle variable-slope concentration−response curves.[8,11] This can occur if the transduction function in the cell converting the receptor stimulus into a tissue response has any element of cooperativity in it to yield concentration−response curves that have slopes different from unity. Therefore, if the cell confers cooperativity on the response-producing elements, then eq 11 can be rewritten as

$$\text{response} = [AR]^n E_{max}/([AR]^n + K_E{}^n) \quad (12)$$

where n is the slope of the concentration−response curve. If the activated receptor is given by the fractional receptor occupancy by the agonist (denoted ρ_A) multiplied by the total receptor concentration $[R_T]$ ($[AR] = \rho_A[R_T]$), then substituting τ for $[R_T]/K_E$ yields

$$\text{response} = \rho_A \tau^n E_{max}/(\rho_A \tau^n + 1) \quad (13)$$

The fractional receptor occupancy ρ_A is given by [active receptor species]^n/[total receptor species]^n. Substituting this definition into eq 13 gives

$$\text{response} = [\text{active receptor species}]^n \tau^n E_{max} / ([\text{active receptor species}]^n \tau^n + [\text{total receptor species}]^n) \quad (14)$$

Defining the concentration of active receptor species in terms of the expression

$$[\text{active receptor species}]^n = [A]^n [R_T]^n / ([A] + K_A)^n \quad (15)$$

and substituting into eq 14 yields the operational model equation for concentration−response curves of variable slope[11]

$$\text{response} = [A]^n \tau^n E_{max}/\{[A]^n \tau^n + ([A] + K_A)^n\} \quad (16)$$

The operational model is the standard for characterizing agonist responses through receptors. The main advantages of the operational model are that it is based on experimental evidence and also furnishes a powerful tool for the prediction of agonism in functional systems. This is because ratios of τ values are system-independent, that is, they relate to the agonist/receptor/signal protein complex and are thus independent of cell type. Figure 3 shows an example of how τ ratios can be used to predict agonism. Two concentration−response curves to agonists 1 and 2 in a functional assay are fit to the model. If a concentration−response curve to one of the agonists is known in any other tissue, the τ ratio can be used to predict the concentration−response curve for the other agonist in that same tissue. In this example, the ratio of efficacies of the two agonists is 25. The concentration−response curves to both

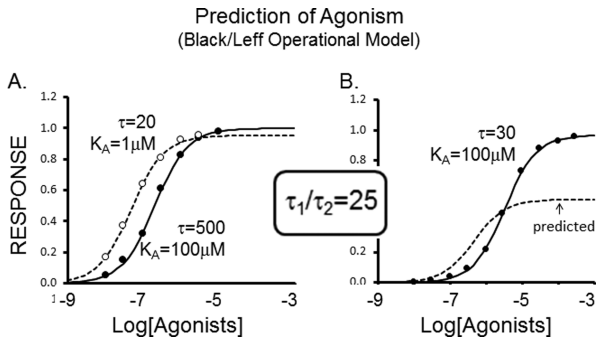

Figure 3. Use of the Black/Leff operational model[8] for predicting agonist response in tissues. (A) In a test system, the model is fit to the data for two agonists: For agonist 2, $\tau = 20$, $K_A = 1$ μM (solid line), and for agonist 1, $\tau = 500$, $K_A = 100$ μM (dashed line). This sets the ratio of efficacies for the agonists at 25. (B) In another less sensitive tissue where the efficacy of agonist 1 is 30, this ratio and the model can be used to predict the curve for agonist 2.

agonists are shown in Figure 3A. In another less sensitive assay system, the efficacy of agonist 1 is 30. By application of the Black/Leff operational equation and the ratio of efficacies ($\tau_1/\tau_2 = 25$), the concentration−response curve for agonist 2 can be predicted (see dotted line, Figure 3B). This procedure can be applied to predict the agonist response in any functional system.

3.2. Series and Parallel Binding Reactions

The operational model was designed to make predictions of effect and simulate agonism, but it does not specifically describe the drug mechanism. Other more complex models have been designed to do this, employing the basic building block of the mass action binding equation (eq 3) with two important variants, specifically the incorporation of two-stage binding (series mass action binding) and the introduction of two receptor states (parallel mass action binding). It is worth considering these processes as additional building blocks for pharmacological models of receptor function.

If the ligand (A) binds to receptors (R) and causes them to change their nature, then series mass action binding occurs. For example, if A binds to R and causes the receptor to convert to the form R*, then the process can be depicted as

$$A + R \overset{K}{\rightleftarrows} AR \underset{\varphi}{\overset{\gamma}{\rightleftarrows}} AR^* \quad (17)$$

The processes controlled by γ and φ will modify the observed affinity of the system for A; therefore, the conformational state of the 7TMR will not be confined to AR but rather will be quite different from the K_A parameter defined in eq 3.[12] With the conversion of the receptor from R to R*, the observed affinity of the ligand for the system will depend on the rate of the transformation to the R* state. The equation for the interaction of A with the complete system is given by[12]

$$K_{obs} = K/[1 + (\gamma/\varphi)] \quad (18)$$

For 7TMRs, such sequential binding reactions involve the interaction of receptors with signaling proteins in the cell membrane such as G-proteins or β-arrestin. Thus, for a signaling element E, the sequential binding is descibed as

$$A + R \overset{K_A}{\leftrightarrows} AR + E \overset{K_E}{\leftrightarrows} ARE \quad (19)$$

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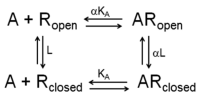
Under these circumstances, the same principles can be applied to yield the overall descriptive constant for the interaction of a ligand A for the entire system as

$$K_{obs} = K_A/(1 + [E]/K_E) \tag{20}$$

It can be seen that the affinity of the ligand for the system will increase in the presence of increasing quantities of signaling molecule E. A general principle introduced by series mass action binding is the fact that the description of the interaction of the ligand with the receptor ceases to be simply a property of the receptor and the ligand but rather becomes a property of the complete system of receptor, interacting molecule, and ligand, that is, the allostesric nature of 7TMRs becomes incorporated into the model (vide infra).

A second very important variant of mass action binding for 7TMRs is the interaction of ligands with more than one state of the receptor R; this comprises a system of parallel mass action binding reactions. A useful model for such interactions is to consider the receptor as an ion channel that can exist in either a closed or an open state, R_{closed} and R_{open}, controlled by an allosteric constant L, which is defined as the ratio $L = R_{open}/R_{closed}$:



[21]

In this type of system, the interaction of A with the complete system does not depend solely upon K_A but rather on the differential activity of the ligand for the two target species, defined as the term α. This is the ratio of the affinity of A for R_{open} to its affinity for R_{closed}

$$K_{obs} = K_A(1 + L)/(1 + \alpha L) \tag{22}$$

It can be seen from eq 22 that the differential affinities of the ligand for the two states and the relative set points of the two receptor species (denoted by L) will control the overall interaction of ligands with the complete system. In fact, if there is a differential affinity of the ligand for the two states (defined as the term α where $\alpha \neq 1$), then the interaction of the ligand with the system necessarily will change the relative proportions of the two states through a process called conformational selection.[13] This can be shown by defining an expression for the ratio of the two states in the absence and presence of a saturating concentration of ligand A as the ratio ρ_∞/ρ_0 (where ρ_0 is the fraction of receptors in the R_{open} state in the absence of ligand and ρ_∞ is the fraction in the presence of a saturating concentration of ligand)

$$\rho_\infty/\rho_0 = \alpha(1 + L)/(1 + \alpha L) \tag{23}$$

If there is a mixture of protein states, then the addition of a ligand with differential affinity will cause a re-equilibration of the mixture according to Le Châtelier's principle (if a chemical system is displaced from equilibrium, changes will act to minimize the deviation from equilibrium) to cause an enrichment of the protein state for which the ligand has the highest affinity. If $\alpha > 1$, the active state (R_{open}) will be enriched, and if $\alpha < 1$, then the inactive state (R_{closed}) will be enriched. This simple idea forms the mechanistic basis for drug efficacy; essentially, drugs will enrich protein species in systems, and the physiological characteristics of those protein species constitute the observed efficacy of the drug.

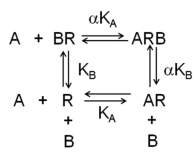
4. SEVEN TRANSMEMBRANE RECEPTORS

Seven transmembrane receptors (7TMRs) are nature's prototypic allosteric proteins in that they reside in cell membranes to form a communication link between extracellular molecules (i.e., neurotransmitters, hormones) and intracellular signaling proteins (G-proteins, β-arrestin, etc.). The communication is achieved through changes in the conformation of the receptor protein or, rather, through the presentation of different combinations of conformations of proteins to the cell after ligand binding through conformational selection. 7TMRs do not stay in static conformations but rather exist in ensembles of different conformations that interconvert in response to the available free energy of the system.[14−19] Equation 23 shows how differential affinity can change the composition of a system with two possible protein states; ensembles can consist of very many more states, and this fact increases the likelihood that drug binding will change the nature of the system. The binding of a saturating concentration of ligand A for a system of 1 to n states as defined by the changes in the amount of a given state i is given by

$$\frac{\rho_\infty}{\rho_0} = \frac{\sum_{i=1}^{n} \alpha_{i+1} L_{i+1}(1 + \sum_{i=1}^{n} L_{i+1})}{(1 + \sum_{i=1}^{n} \alpha_{i+1} L_{i+1}) \sum_{i=1}^{n} L_{i+1}} \tag{24}$$

Only in the case where the ligand recognizes every single state as being identical (where $\alpha_1 = \alpha_2 = \alpha_3 = \cdots = \alpha_i = 1$) will there be no change in the composition of the ensemble. This leads to the prediction that ligand binding will not be a passive process (i.e., will not change the nature of receptor ensembles) but rather will change synoptic receptor systems. The corollary to this idea is that efficacy (the ability of ligands to change receptors) is probably far more prevalent a drug property than previously thought when the functional assay systems available to measure receptor function were restricted to few in number.

The two building blocks defined by eqs 19 and 21 can be amalgamated into a more complex building block for pharmacological receptor systems typified by the so-called "allosteric binding equation". 7TMRs are allosteric in nature, as they form a conduit for energy transfer (an "allosteric vector"[20]), between a modulator and a guest molecule both binding to the receptor. The activities of both the modulator and guest are conditional upon the binding of the other species. This more complex building block is composed of a collection of series and parallel mass action binding proecesses[21,22]



[25]

where the modulator is B, the receptor is R, and the guest molecule is A. The allosteric energy vector is bidirectional; therefore, the affinity of the modulator is modified according to the parameter α, and the quantity of guest is modified by the factor α and the quantity of modulator. Therefore, the interaction of the modulator with the complete system is given by

$$K_{obs} = K_A(1 + [B]/K_B)/(1 + \alpha[B]/K_B) \tag{26}$$

The reciprocation of the effects of modulator and guest has been verified in numerous 7TMR systems, including X-ray crystallography of the 7TMR structure through differences in the structure of the β_2-adrenoceptor bound and not bound to a

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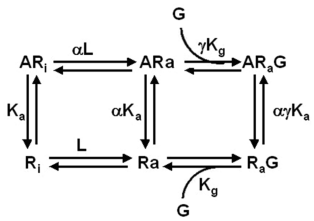
nanobody simulating a G-protein.[23,24] Similarly, changes in the ghrelin receptor conformation in lipid disks are observed upon addition of Gq protein and β-arrestin cobinding to the receptor.[25] In addition, large (18-fold) changes in the affinity of a guest ligand, in this case, the κ-opioid receptor agonist salvinorin, mediated by concentrations of modulator $G_{\alpha 16}$ and/or $G_{\alpha i2}$ proteins (according to eq 26) have been demonstrated using the substituted cysteine accessibility method.[26] In general, the behavior of ligands as they cause changes in the receptor species in live cell membranes has been termed "receptor distribution"[27] and plays an important role in the overall behavior of pharmacological receptor systems.

4.1. Ternary Complex Models

More detailed models of 7TMR function involve combinations of the various building blocks described previously. The first mechanistic model of 7TMR function that emerged from experimental ideas involving mobile receptors and ternary complex formation[28] was the ternary complex model of 7TMR function. This model evolved from the finding that guanine nucleotides could affect the affinity of agonists through changes in the interaction of receptors and G-proteins. This suggested two-stage binding of ligand to receptor and, subsequently, complexation with a G-protein.[29−31] These ideas culminated in a model termed the "ternary complex model" by DeLean and colleagues.[32] The ternary complex model is depicted in eq 25, where the G-protein is the molecular species denoted as B.

A major experimental finding that required changes to the ternary complex model was the observation of constitutive receptor activity. Specifically, the ternary complex model has no provision for the activation of receptors in the absence of ligand. However, Costa and Herz[33] showed definitively that receptors could spontaneously exist in an active state in the absence of agonist ligand. Therefore, the ternary complex model was modified to accommodate this experimental finding by the proposal that receptors spontaneously can exist in an activated form. The modification resulted in the extended ternary complex model,[34] which describes the spontaneous formation of an active-state receptor ($[R_a]$) from an inactive-state receptor ($[R_i]$) according to an allosteric constant ($L = [R_a]/[R_i]$). The active-state receptor can form a complex with G-protein ($[G]$) spontaneously to form R_aG, or agonist activation can induce formation of the ternary complex AR_aG

In terms of eq 27, the fraction ρ of G-protein activating species (producing response), namely, $[R_aG]$ and $[AR_aG]$, as a



[27]

fraction of the total number of receptor species, $[R_{tot}]$, is given by

$$\rho = \frac{L[G]K_G(1 + \alpha\gamma[A]/K_A)}{[A]/K_A\{1 + \alpha L(1 + \gamma[G]/K_G)\} + L(1 + [G]/K_G) + 1} \quad (28)$$

where $[A]$ represents the ligand concentration and K_A and K_G are the equilibrium dissociation constants of the ligand−receptor and G-protein−receptor complexes, respectively. The difference in the affinity of the ligand for R_a over R_i is given by α; for example, for $\alpha = 5$, the ligand has a 5-fold greater affinity for R_a than for R_i. Similarly, the term γ defines the multiple difference in affinity of the receptor for the G-protein when the receptor is bound to the ligand. Thus, $\gamma = 20$ means that the ligand-bound receptor has a 20-fold greater affinity for the G-protein than the ligand-unbound receptor.

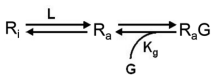
The constants α and γ quantify the ability of the ligand to selectively cause the receptor to couple to G-proteins, and as such, they manifest ligand efficacy. For example, if a ligand produces a bias of the system toward more active receptor species (positive α) and/or enables the ligand-occupied receptor to bind to G-proteins with a higher affinity (positive γ), then it will be an agonist with positive efficacy. Similarly, if a ligand selectively stabilizes the inactive state of the receptor ($\alpha < 1$) or reduces the affinity of the receptor for G-proteins ($\gamma < 1$), then it will reduce the levels of spontaneously active receptor and reverse the elevated basal receptor activity.

As discussed previously, the extended ternary complex model was developed to account for the phenomenon of constitutive receptor activity. This phenomenon was first observed in certain functional systems where an apparent basal level of activity was further reduced by certain ligands in a concentration-dependent manner. These ligands were labeled inverse agonists, as they caused a decrease in the level of a spontaneously formed active receptor species that was responsible for cellular signaling. One of the earliest examples of this was the opioid ligand ICI174 in NG-108 cells containing naturally expressed μ-opioid receptors.[33] An important requirement of a functional system to show constitutive receptor activity is the availability of a signaling protein for interaction with the spontaneously active receptor species; this is what constitutes the reporting system for constitutive receptors. For example, considering a system of activated receptors R_a and G-proteins according to eq 27, constitutive receptor activity (as defined by the presence of the spontaneously signaling species $[R_aG]$) is given by

$$\rho = L[G]/K_G/\{L(1 + [G]/K_G) + 1\} \quad (29)$$

Equation 29 shows that constitutive activity can be produced by high G-protein concentration, high-affinity receptor−G-protein coupling (low value of K_G), and/or a high tendency for the receptor to spontaneously form the active state (variable values of L).

Before the discovery of constitutive G-protein-coupled receptor (GPCR) activity, efficacy was considered to be only a positive vector, that is, the ability of a ligand to promote receptor−signaling protein interactions. The observation of reductions in spontaneous receptor activity by ligands clearly indicated that efficacy is a vector with negative, as well as positive, values. A common method of producing constitutively active functional assay systems is through the overexpression of



[30]

receptors. Considering the schemethe dependence of constitutive activity on $[R_i]$ is given by

$$[R_aG]/[G_{tot}] = [R_i]/\{[R_i] + (K_G/L)\} \quad (31)$$

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where $[R_i]$ is the receptor density, L is the allosteric constant describing the propensity of the receptor to spontaneously adopt the active state, and K_G is the equilibrium dissociation constant for the activated receptor−G-protein complex. Constitutive activity is favored by a low energy barrier to spontaneous formation of the active state (large values of L) and/or a tight coupling between the receptor and the G-protein (low value of K_G).

The extended ternary complex model is thermodynamically incomplete in that it does not allow the interaction of the inactive receptor species with the signaling species (i.e., G-protein). This theoretical shortcoming is eliminated in the cubic ternary complex model;[35−37] see Figure 4. This allows ligands to form inactive complexes and sequester signaling proteins; that is, an inverse agonist (one that stabilizes the inactive state of the receptor) theoretically can form inactive ternary complexes and thus sequester G-proteins away from signaling pathways. This, in fact, has been shown to be the case with cannabinoid receptors.[38] In the cubic ternary complex model, the two receptor species ($[R_a]$ and $[R_i]$) can form complexes with the signaling protein; if the signaling protein is a G-protein, then the respective species are $[R_aG]$, which will produce a physiological signal, and $[R_iG]$, which will not. In the presence of a ligand [A], the corresponding species are $[AR_aG]$ and $[AR_iG]$.

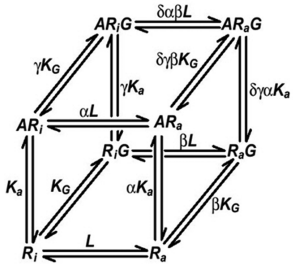

Figure 4. Components of the cubic ternary complex model.[35−37] The major difference between this model and the extended ternary complex model[34] is the potential for formation of the $[AR_iG]$ complex and the $[R_iG]$ complex, both receptor/G-protein complexes that do not induce dissociation of G-protein subunits and subsequent response. The efficacy terms in this model are α, γ, and δ.

Referring to Figure 4, the equilibrium equations for the various species are

$$[AR_i] = [AR_aG]/\alpha\gamma\delta\beta L[G]K_g \tag{32}$$

$$[AR_a] = [AR_aG]/\gamma\beta\delta[G]K_g \tag{33}$$

$$[R_a] = [AR_aG]/\alpha\gamma\delta\beta[G]K_g[A]K_a \tag{34}$$

$$[R_i] = [AR_aG]/\alpha\gamma\delta\beta L[G]K_g[A]K_a \tag{35}$$

$$[R_aG] = [AR_aG]/\alpha\gamma\delta[A]K_a \tag{36}$$

$$[R_iG] = [AR_aG]/\alpha\gamma\delta\beta L[A]K_a \tag{37}$$

$$[AR_iG] = [AR_aG]/\alpha\delta\beta L \tag{38}$$

The conservation equation for receptor species is

$$[R_T] = [AR_aG] + [AR_iG] + [R_iG] + [R_aG] + [AR_a] + [AR_i] + [R_a] + [R_i] \tag{39}$$

It is assumed that the physiological response emanates from complexes formed with the active receptor species R_a ($[AR_aG]$ and $[R_aG]$). The fraction of the response-producing species among the total receptor species $\{([AR_aG] + [R_aG])/R_T\}$ is denoted ρ and is given by

$$\rho = \frac{\frac{\beta L[G]}{K_G}(1 + \alpha\gamma\delta[A]/K_A)}{\frac{[A]}{K_A}\{1 + \alpha L + \gamma[G]/K_G(1 + \alpha\beta\delta L)\} + \frac{[G]}{K_G}(1 + \beta L) + 1} \tag{40}$$

In contrast to the extended ternary complex model, the cubic ternary complex model predicts that the constitutive activity of receptor systems can reach a maximal asymptote that is below the system maximum (partial constitutive activity). This is caused by the fact that the cubic ternary complex model predicts the maximal constitutive activity is given by (see eq 40 where $[A] = 0$ and $[G] \to \infty$):

$$\text{maximal constitutive activity} = \beta L/(1 + \beta L) \tag{41}$$

From eq 41, it can be seen that the constitutive activity need not reach a maximal asymptote of unity. The cubic ternary complex is heuristic in that there are more individually nonverifiable constants than other models, and this makes it limited in practical application. However, this model does subsume many other receptor models in the pharmacological literature; Figure 5 shows a number of models proposed as facets of the cubic configuration.

5. ALLOSTERIC NATURE OF 7TMRS

Before evidence showed that receptors coupled to other membrane proteins can elicit effects (i.e., the ternary, extended ternary, and cubic ternary complex models), receptors were

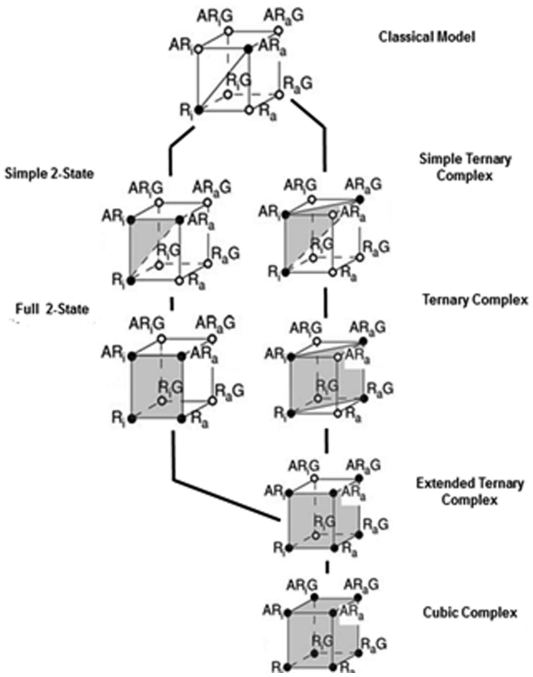

Figure 5. 7TMR receptor models as various parts of the thermodynamically complete cubic ternary complex model: classical model,[2,4,7,39−41] simple two-state model,[42−46] simple ternary complex model,[47−52] full two-state model,[45,53−56] ternary complex model,[57−64] extended ternary complex model,[35,65] cubic ternary complex model.[35−37]

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basically thought to be switches. With further experimentation, it became evident that a more complex binding model, such as that shown in eq 25, was operative for 7TMRs. The fact that a ternary species such as [ABR] is formed highlighted the allosteric nature of GPCRs. It is worth considering receptors in allosteric terms in order to develop the current model of functional allostery for receptors.

The word *allosteric* comes from the Greek *allos*, meaning different, and *steric*, which refers to arrangement of atoms in space; the word *allostery* refers to the property of changing shape. Allosteric effects were first described for enzymes by pioneers such as Koshland.[65] The fact that metabolic enzymes could be inhibited by structurally diverse products of subsequent enzymes in a metabolic pathway led to the postulate that proteins can be modulated through binding of molecules at more than one location.[66] In the case of receptors, the idea is that the interaction of a receptor protein with other bodies such as ligands and cellular signaling proteins causes the receptor to change its shape (more specifically, to be stabilized toward certain conformations). The interactions of the receptor with various bodies occur at differing locations on the receptor protein; thus, the interactions of species such ligands and receptors and signaling proteins and receptors occur through changes in the shape of the protein. As ligands for receptors were discovered that did not conform to conventional orthosteric models of receptor function, it became apparent that receptor effects were allosteric. Various techniques, notably point mutation, show that some pharmacological antagonists bind to sites distinct from those utilized by the endogenous agonist (i.e., hormone, neurotransmitter) to alter binding and subsequent tissue response. For example, point mutations of muscarinic m1 receptors show that substitution of an aspartate residue at position 71 affects the affinity of the allosteric modulator gallamine but not the affinity of the competitive radiolabeled antagonist [^{3}H]-*N*-methylscopolamine.[67] For enzymes, it was shown that some allosteric sites are separated by large molecular distances, as in the case of glycogen phosphorylase *b*, where the binding site for the allosteric modulator CP320626 is 33 Å from the catalytic site and 15 Å from the site for the substrate cyclic adenosine monophosphate (AMP).[68]

In terms of allosteric transitions of protein conformations (i.e., changes in conformation due to the binding of allosteric ligands), it is useful to consider a binding ensemble model of receptors and the mechanism of conformational selection (eq 24). Thus, an allosteric modulator could have a high affinity for some of the naturally occurring receptor states and selectively stabilize them upon binding. These preferred states might otherwise be quite rare in the natural library of conformations exhibited by the receptor, leading to a preponderance of a dominant receptor species with completely different properties with respect to the binding of ligands and signaling proteins. The fact that these conformations might be rare could lead to a requirement of longer equilibration times for complete binding of allosteric ligands to receptors at allosteric sites, a fact supported by some observations. Thus, the allosteric p38 mitogen-activated protein (MAP) kinase inhibitor BIRB 796 {1-(5-*tert*-butyl-2-*p*-tolyl-2H-pyrazol-3-yl)-3-[4-(2-morpholin-4-yl-ethoxy)naphthalen-1-yl]urea} binds to a conformation of MAP kinase requiring movement of a Phe residue by 10 Å (so-called "out" conformation) with an association rate of 8.5×10^5 M^{-1} s^{-1}, 50 times slower than required for other inhibitors (4.3×10^7 M^{-1} s^{-1}). The result is that, whereas other inhibitors reach equilibrium within 30 min, BIRB 796 requires a full 2 h of equilibration time.[69] The fact that a completely new receptor species could result from the binding of an allosteric ligand causes allosteric ligands to have unique properties when compared to conventional orthosteric ligands. Before these species are discussed, it is useful to derive the current model for functional allosterism for 7 TMRs.

5.1. Functional Allosteric 7TMR Model

The allosteric binding model is formally identical to the scheme shown in eq 25 with the following definitions. A probe ligand, denoted A, is defined that reports a certain receptor state; this is usually an agonist for functional studies or a radioligand in binding studies. A second ligand, denoted B, is defined that can bind to a separate site on the receptor; thus, there can be a receptor species with both ligands bound (ABR). The equilibrium association constants for the binding of each ligand to form receptor association complexes are denoted as K_a and K_b, respectively. As shown in eq 25, the binding of either ligand to the receptor modifies the affinity of the receptor for the other ligand by a factor of α. The resulting equilibrium equations are

$$K_a = [AR]/[A][R] \tag{42}$$

$$K_b = [BR]/[B][R] \tag{43}$$

$$\alpha K_a = [ARB]/[BR][A] \tag{44}$$

$$\alpha K_b = [ARB]/[AR][B] \tag{45}$$

Solving for the agonist-bound receptor species [AR] and [ARB] as a function of the total receptor species ($[R_{tot}] = [R] + [AR] + [BR] + [ARB]$) yields

$$[AR] + [ARB] = \frac{((1/\alpha[B]K_b) + 1)}{((1/\alpha[B]K_b) + (1/\alpha K_a) + ((1/\alpha[A]K_aK_b) + 1)} \tag{46}$$

Converting association to dissociation constants (i.e., $K_A = 1/K_a$) yields

$$[AR]/[R_{tot}] = \frac{[A]/K_A(1 + \alpha[B]/K_B)}{[A]/K_A(1 + \alpha[B]/K_B) + [B]/K_B + 1} \tag{47}$$

This is the allosteric binding equation.[21,22] The parameter α defines the change in the affinity of A for the receptor in the presence of bound allosteric ligand B. For instance, a value for α of 0.03 means that the allosteric antagonist causes a 30-fold reduction in the affinity of the receptor for the agonist. This can be seen from the relationship describing the affinity of probe A for the receptor in the presence of varying concentrations of antagonist

$$K_{obs} = K_A(1 + [B]/K_B)/(1 + \alpha[B]/K_B) \tag{48}$$

This binding model can then be extended to functional systems by allowing the ligand-bound species AR, BR, and ABR to elicit responses through a mechanism in the cell by linking the allosteric binding model to the Black/Leff operational model for receptor function

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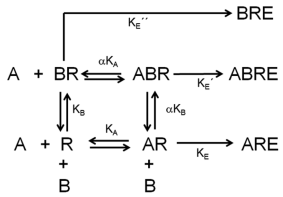

[49]

For this scheme, the concentrations of the additional equilibrium species are given by

$$[\mathrm{AR}] = [\mathrm{ABR}]/\alpha[\mathrm{B}]K_\mathrm{b} \tag{50}$$

$$[\mathrm{BR}] = [\mathrm{ABR}]/\alpha[\mathrm{B}]K_\mathrm{b} \tag{51}$$

$$[\mathrm{R}] = [\mathrm{ABR}]/\alpha[\mathrm{B}]K_\mathrm{b}[\mathrm{A}]K_\mathrm{a} \tag{52}$$

The operational model dictates that the response-producing species activate the response elements of the cell according to

$$\text{response} = ([\mathrm{AR}]/K_\mathrm{E} + [\mathrm{ABR}]/K'_\mathrm{E}) / ([\mathrm{AR}]/K_\mathrm{E} + [\mathrm{ABR}]/K'_\mathrm{E} + 1) \tag{53}$$

where K_E and K'_E refer to equilibrium dissociation constants of the agonist ligand and modulator ligand receptor complexes, respectively, as they produce a cellular response. Redefining the amount of any receptor species as the fractional amount of receptor multiplied by the total receptor number, eq 53 can be rewritten as

$$\text{response} = \frac{\rho_\mathrm{AR}[\mathrm{R_T}]/K_\mathrm{E} + \rho_\mathrm{ABR}[\mathrm{R_T}]/K'_\mathrm{E}}{\rho_\mathrm{AR}[\mathrm{R_T}]/K_\mathrm{E} + \rho_\mathrm{ABR}[\mathrm{R_T}]/K'_\mathrm{E} + 1} \tag{54}$$

where the fraction of receptors in AR form, ρ_AR, is given by

$$\rho_\mathrm{AR} = \frac{[\mathrm{A}]/K_\mathrm{A}}{[\mathrm{A}]/K_\mathrm{A}(1 + \alpha[\mathrm{B}]/K_\mathrm{B}) + [\mathrm{B}]/K_\mathrm{B} + 1} \tag{55}$$

Similarly, the fraction of receptors in ABR form, ρ_ABR, is given by

$$\rho_\mathrm{ABR} = \frac{(\alpha[\mathrm{A}]/K_\mathrm{A})([\mathrm{B}]/K_\mathrm{B})}{[\mathrm{A}]/K_\mathrm{A}(1 + \alpha[\mathrm{B}]/K_\mathrm{B} + [\mathrm{B}]/K_\mathrm{B} + 1)} \tag{56}$$

Through the operational model,[8] the efficacy of agonist A is given by $\tau_\mathrm{A} = [\mathrm{R_T}]/K_\mathrm{E}$ in the absence of the modulator [B] and $\beta\tau_\mathrm{A}$ in the presence of the modulator, where $\beta\tau_\mathrm{A} = [\mathrm{R_T}]/K'_\mathrm{E}$. Substituting eqs 54 and 55 into eq 53 yields[70−72]

$$\text{response} = \frac{\frac{\tau_\mathrm{A}[\mathrm{A}]}{K_\mathrm{A}}\left(1 + \frac{\alpha\beta[\mathrm{B}]}{K_\mathrm{B}}\right) + \frac{\tau_\mathrm{B}[\mathrm{B}]}{K_\mathrm{B}}}{\frac{[\mathrm{A}]}{K_\mathrm{A}}\left\{1 + \frac{\alpha[\mathrm{B}]}{K_\mathrm{B}} + \tau_\mathrm{A}\left(1 + \frac{\alpha\beta[\mathrm{B}]}{K_\mathrm{B}}\right)\right\} + \frac{[\mathrm{B}]}{K_\mathrm{B}}(1 + \tau_\mathrm{B}) + 1} \tag{57}$$

5.2. Properties of Allosteric Ligands

Equation 57 describes all possible effects of an allosteric modulator on a receptor system, namely, direct production of receptor response through efficacy τ_B, alteration of the affinity of the primary agonist through α, and alteration of the efficacy of the primary agonist through β. This model is capable of describing any changes to an agonist concentration−response curve, but in general, there are three phenotypic effects of allosteric ligands with two variants that are seen experimentally; these are shown in Figure 6. Because allostery essentially produces a new receptor, there are no rules to link what combinations of changes in the efficacy and affinity will be produced; that is, α and β are independent parameters and can

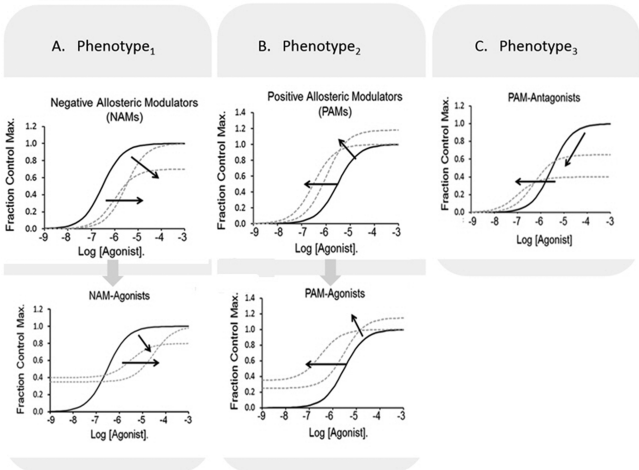

Figure 6. General phenotypes of effects of allosteric molecules on agonist concentration−response curves. Allosteric molecules can increase, not change, or decrease affinity and efficacy, and these effects, in various combinations, can produce dextral or sinistral shifts of the curves and increases or decreases in the maximal response. In addition, the allosteric modulator might exhibit direct efficacy to cause agonist response in the system.

change in opposite directions. One phenotype is a general diminution of agonism, and the modulators causing this are referred to as negative allosteric modulators (NAMs; see Figure 6A). The observed effect on agonist concentration−response curves can be dextral displacement of the curve with or without concomitant depression of maximal response. These effects are produced by modulators with $\alpha < 1$, $\beta < 1$, or some combination in which the product satisfies $\alpha\beta < 1$. A variant of this profile is the NAM agonsits, for which $\tau_\mathrm{B} > 1$ (see Figure 6A). Another allosteric phenotype provides a modulator-induced increase in the agonist response; these molecules are referred to as positive allosteric modulators (PAMs; see Figure 6B). PAMs produce sinistral displacements of concentration−response curves and/or increased maximal responses. These molecules have $\alpha > 1$ and/or $\beta > 1$ or some combination in which the product satisfies $\alpha\beta > 1$. A variant of this phenotype occurs when direct agonism ($\tau_\mathrm{B} > 1$) is concomitant with PAM activity; see Figure 6B.

As discussed previously, allosteric modulators can have separate effects on agonist affinity and efficacy, leading to the possibility that an allosteric modulator could change agonist affinity in one direction and efficacy in another. For example, the CCR5 receptor allosteric modulator aplaviroc blocks the physiological response to the chemokine CCL5 but has no effect on the binding of CCL5 to the receptor.[73,74] Similarly, the ester CPCCOEt (7-hydroxyiminocyclopropan[b]chromen-1a-carboxylic acid ethyl ester) completely blocks the responses to glutamate in CHO cells naturally expressing human GluR1b receptors but has no effect on glutamate binding.[75] This can lead to a unique phenotype referred to as PAM antagonism seen with allosteric molecules that increase the affinity of the receptor for the agonist ($\alpha > 1$) but prevent the agonist from producing cellular activation ($\beta < 1$). In the presence of PAM antagonists, the concentration−response curves for agonists shift to the left, but the maximal responses are depressed (see Figure 6C). Interestingly, this confers a unique property on PAM antagonists in that they become more potent with increasing agonism. This is due to the fact that the energy

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transfer from the modulator to the agonist in the form of α is reciprocal; that is, just as the modulator increases the affinity of the receptor for the agonist, so too does the agonist increase the affinity of the receptor for the modulator. Reciprocal effects for allostery have been shown in binding experiments with radiolabeled probes and allosteric molecules.[76] Thus, the potency of the PAM antagonist ifenprodil as a blocker of the effects of the agonist *N*-methyl-D-aspartate (NMDA) increases 10-fold when the concentration of NMDA to be blocked is increased 10-fold.[77] To date, only a small number of molecules have been shown to be PAM antagonists, specifically for NMDA receptors,[77] cannabinoid 1 receptors,[72] and FFA-3 receptors.[78]

The functional allosteric model described by eq 57 predicts some unique features of allosteric molecules that extend to their possible therapeutic application. The first of these is saturability of effect, which naturally results from the fact that the allosteric effect will reach a maximal asymptote as the allosteric site becomes fully occupied by the modulator. This is in contrast to simple competitive antagonists, for which the degree of antagonism theoretically is infinite for an infinite concentration of antagonist. In terms of binding affects, the maximal change in affinity that can be produced by the allosteric modulator is $K_{obs}/K_A = K_A/\alpha K_A = \alpha^{-1}$. The separation of binding loci also dissociates the length of time for allosteric effect from its magnitude. Specifically, a longer time of action (target coverage) can be achieved for an orthosteric antagonist by increasing the in vivo concentration, but this will necessarily be linked to a larger magnitude of effect. In contrast, allosteric antagonists do not have this property because the magnitude of antagonism is limited by the allosteric term α. Therefore, after saturation of the allosteric binding site, further increases in concentration serve only to increase the duration of action, not the magnitude of effect.

The second important property of allosteric systems is linked to the fact that allosteric effects are mediated by a change in the conformation of the receptor protein. Under these circumstances, there is no a priori reason that a change in the shape of the protein (which affects the actions of one receptor probe, i.e., the agonist) should be the same as the change in effect of another receptor probe; this property is referred to as probe dependence. For example, the CCR5 allosteric modulator aplaviroc produces very little effect on the binding of chemokine CCL5 to the receptor but completely blocks the binding of chemokine CCL3.[74] Probe dependence has broad implications for the application of allosteric modulators to therapeutics, as well as serious implications for how allosteric drugs are detected. For instance, a PAM for the natural neurotransmitter acetylcholine theoretically would be a useful therapy in Alzheimer's disease, where there is a failure of cholinergic neuronal transmission.[79,80] A problem arises in the drug screening process for such drugs. For example, the natural agonist neurotransmitter acetylcholine is chemically unstable and unsuitable for use as a receptor probe in high-throughput screens; for this reason, surrogate cholinergic agonists such as carbachol and pilocarpine have been used. However, allosteric probe dependence can obfuscate screening data as in the case of the PAM LY2033298, which causes agonist-dependent differential potentiation of different agonists such as acetylcholine and oxotremorine.[81] Therefore, these screens might detect PAMs for the surrogate probe that might not then produce potentiation of the natural agonist acetylcholine. Probe dependence can also be an issue for receptors with more than one natural ligand. The potential antidiabetic PAM NOVO2 produces a 25-fold potentiation of a minor natural agonist oxyntomodulin for the GLP-1 receptor but only a 1.5-fold potentiation of the main natural agonist GLP-1(7−36)NH_2.[82] Probe dependence can also be exploited for therapeutic advantage as in the case of allosteric HIV-1 entry inhibitors for the treatment and prevention of AIDS. Thus, whereas HIV-1 utilizes the CCR5 chemokine receptor to gain entry into cells to cause infection, natural CCR5 receptor function (interaction with natural chemokines) is beneficial in preventing progression from HIV-1 infection to AIDS.[83] Therefore, a probe-dependent CCR5 allosteric modulator (one that blocks HIV-1 interaction but otherwise spares natural CCR5-chemokine interaction) would give an added therapeutic advantage. Interestingly, of the known allosteric HIV-1 entry inhibitors interacting with CCR5, there is a 500-fold variation in the relative ability to block HIV-1 infection vs CCR5 function.[84]

Because allosteric receptor effects have the ability to produce global conformational changes, numerous regions of interaction of these proteins are changed through allosteric effects. This leads to a useful therapeutic property of allosteric ligands, namely, the ability to cause changes in the interactions of very large proteins. For example, the allosteric modulator aplaviroc can block the interaction of two very large proteins, namely, the chemokine CCR5 receptor and the HIV-1 viral coat protein gp120 to prevent HIV-1 infection of cells.[74] Allosterically induced global conformational changes in receptors can also confer a therapeutically useful property on allosteric antagonists, namely, texture in antagonism. Whereas orthosteric antagonists binding to receptors lead to a common end product upon saturation of binding, modulator-induced conformational changes need not be identical, and different allosteric modulators can produce pharmacological blockades through the production of different receptor conformational species. This can have therapeutic implication as in the blockade of HIV-1 entry in the therapy for AIDS in terms of overcoming viral resistance with chronic therapy. In AIDS, this issue might be of importance, as it is expected that HIV-1 viral mutation will eventually lead to tolerance to an HIV-1 entry inhibitor because the virus changes to a form that can utilize the allosterically modified receptor.[85,86] The ability to produce different conformations of the CCR5 receptor would be a useful feature to overcome expected viral resistance in this disease. Interestingly, antibody binding profiles of two antibodies for the CCR receptor, Ab45531 and Ab45523, have shown that the allosteric modulators TAK779, SCH-C, and aplaviroc produce different tertiary conformations of the CCR5 receptor.[87] This leads to the possibility that serial therapy with different allosteric modulators might be a strategy for overcoming viral resistance.

Another property of allosteric modulators is their permissive quality; that is, because they bind to separate loci on the receptor protein, there is always the potential that the natural agonist will still interact with the receptor and produce an effect. This also gives allosteric modulators the ability to modify or modulate natural agonist signaling without precluding it (hence the term "modulator"). In addition, allosteric modulators have the potential to preserve complex physiological patterns of innervation and structure. This can be especially useful for therapeutic PAMs, as these molecules can induce a physiological effect that preserves the physiological "wiring" of complex systems (i.e., in failing neuronal systems

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such as Alzheimer's disease, the circuitry of the brain must be preserved). PAMs can also reduce desensitization and side effects because no effect is produced until the system becomes activated with natural agonist. Finally, extraordinary selectivity can be achieved with allosteric modulators in contrast to orthosteric ligands, which bind to the natural agonist binding site on receptors. This is because natural binding sites can be highly conserved between receptor types due to the fact that they must recognize common ligands. Such conservation is not necessary for allosteric binding sites, and allosteric binding loci can be more diverse between subtypes of a receptor.[88−92]

When considering 7TMRs, it is useful to introduce allosteric nomenclature and also to define the vectorial nature of allostery. Thus, an allosteric vector is defined as the binding of a modulator to the receptor protein as a conduit to cause a change in the interaction of the conduit toward a guest molecule (which could be a ligand, other receptor, or cytosolic signaling protein). As discussed for PAM antagonists, the energy of the allosteric vector travels both ways; therefore, a modulator could just as well be considered the guest, and the guest, the modulator. Using a nomenclature in which the allosteric vector is directed from the modulator through the conduit to the guest, there are three basic orientations in receptor pharmacology that are relevant to 7TMR function. These are (1) guest allostery (Figure 7A), whereby two ligands

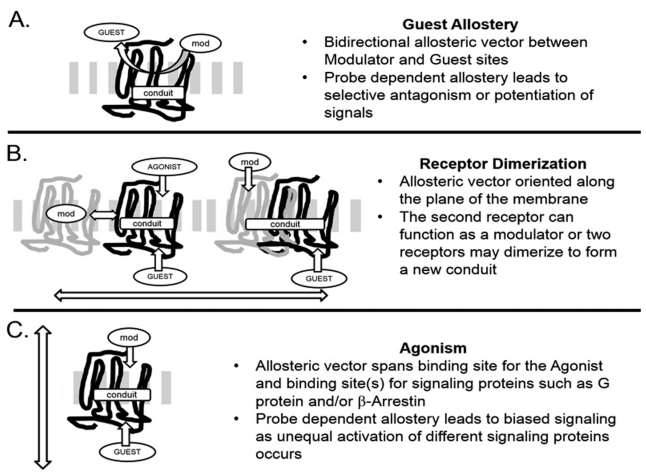

Figure 7. Orientations of the allosteric vector from a modulator, through a conduit, toward a guest. (A) Guest allostery: The modulator affects the interaction of the receptor with a guest molecule. (B) Receptor dimerization: When the allosteric vector is oriented along the plane of the cell membrane (oligomerization), another receptor or membrane protein can either function as a modulator or form a new conduit to allow a new range of guest interactions to occur. (C) Agonism: The allosteric vector is oriented toward the cell cytosol, where the modulator changes the interaction of cellular proteins with the receptor; this usually results in agonism.

bind at different locations on the receptor protein and the binding of each affects the binding and function of the other; (2) receptor dimerization, in which the vector is directed along the plane of the membrane where two receptors (or more) or other membrane-bound proteins form a complex with the receptor (Figure 7B); and (3) agonism, in which the vector is directed toward the cytosol (agonism) where a ligand affects the interaction of the receptor with a signaling protein in the cytosol[20] (Figure 7C). It is worth considering each of these mechanisms in turn.

Guest allostery historically was the first phenomenon to be recognized and quantified. This effect is explicitly described by eq 25, where the probe molecule (agonist) is A and the allosteric guest molecule is B; in terms of allosteric reciprocity, the agonist can be considered the guest and molecule B the modulator (i.e., functions as a PAM or NAM). The allosteric vector can also be oriented along the plane of the cell in systems where two or more receptors produce receptor oligomers.[93−95] Alternatively, the interactions can involve a receptor and/or other membrane-bound protein such as a receptor-activity-modifying protein (RAMP).[96] Cell surface oligomerization can result in two types of system: (1) a receptor that can function as a modulator to modify an existing agonist/receptor/G-protein system or (2) a receptor that can dimerize to form a new conduit and thus create a system whereby a ligand binding to one of the receptors can affect the signaling controlled by the other receptor (Figure 7B). Reviews on these often very complex systems have been published in the literature; see refs 97−101. The allosteric vector can also be directed toward the cell cytosol; in these systems, the modulator controls the interaction of the receptor with signaling proteins such as G-proteins, GRKs, β-arrestin, and other coupling proteins and can thus function as an agonist; see Figure 7C. In this type of system, the phenomenon of allosteric probe dependence is particularly important from a therapeutic point of view. Specifically, it is known that different modulators (agonists) stabilize different tertiary receptor conformations that have differential affinities for signaling proteins. This can cause them to selectively activate certain signaling pathways in a cell at the expense of others, that is, to produce biased signaling.[102−107] There are indications that biased ligands can be superior therapeutic entities for the treatment of congestive heart failure,[108,109] psychosis,[110] pain,[111−113] osteoporosis,[114−116] and diabetes.[117]

There are methods based on the Black/Leff operational model[8] to quantify the degree of bias for any given molecule. Specifically, an index of agonism is obtained for each signaling pathway in the form of $\log(\tau/K_A)$ values;[118] within any given signaling pathway, ratios of test agonist values and a reference agonist are obtained to yield $\Delta\log(\tau/K_A)$ values, and then ratios of the ratios are obtained as measures of bias in the form of $\Delta\Delta\log(\tau/K_A)$ values (see Figure 8). An example of this procedure is given in Table 1 for dopamine receptor agonists.[119] In this case, it can be seen that, even though both agonists quinpirole and FAUC321 cause elevation of cyclic AMP and stimulation of phosphor-ERK signaling, FAUC321 produces a 6.02-fold greater pERK phosphorylation response per given unit of cyclic AMP elevation than does quinpirole, that is, FAUC321 is 6-fold-biased toward the pERK signaling pathway.

5.3. Extension of the Bias Model to Allosteric Molecules

Allosteric molecules are permissive because they bind to a site separate from the natural ligand binding site. Under these circumstances, the natural agonist can bind and have the potential to signal even in the presence of the allosteric modulator; that is, the natural signal might be blocked, modulated, or enhanced, or the quality of the signal might change. With the advent of biased signaling and the discovery that agonists cause pleiotropic signaling (i.e., have many efficacies linked to various signaling pathways), any bias between agonists will result in a change in the overall amalgamated cell signal; that is, the quality of the efficacies

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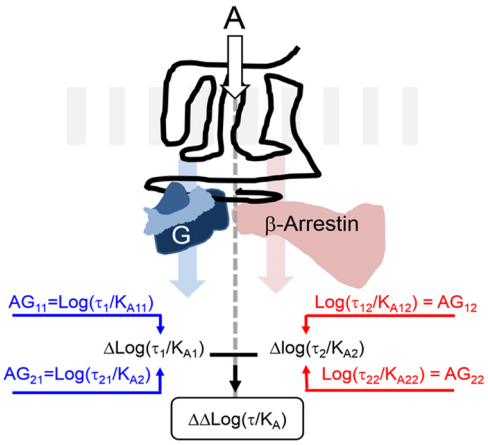

Figure 8. Biased signaling: The interactions of agonists are quantified with the Black/Leff operational model[8] to furnish characteristic $\log(\tau/K_A)$ values for two signaling pathways. Within each pathway, the relative powers of these agonists are quantified through the calculation of $\Delta\log(\tau/K_A)$ values. If these relative measures of activity within each pathway are made to a common reference agonist, the $\Delta\log(\tau/K_A)$ values can then be used to calculate $\Delta\Delta\log(\tau/K_A)$ values, the antilogarithms of which represent the biases.

for those agonists will be different. Therefore, allosteric molecules have the potential to change the quality of the efficacy of natural agonists, and this can be therapeutically important. For example, the natural agonist neurokinin A interacts with NK2 receptors to activate G_s and G_q, but upon binding of NAM LP1805, this changes to a pattern of enhanced G_q activation and blockage of G_s activation.[120] Similarly, the CRH2 receptor natural agonist prostaglandin D_2 signaling profile, normally a dual activation of G_i and β-arrestin, is changed by the NAM *N*-tosyltryptophan to a profile of sole activation of G_i protein (with no concomitant β-arrestin interaction).[121] The same biasing effects can be seen with PAMs. For example, the GLP-1 natural agonist GLP-1(7−36)NH_2 produces elevated cyclic AMP, calcium, and phospho ERK effects. In the presence of the PAM NOVO2, the cyclic AMP response of this natural agonist is potentiated, but very little effect is seen on calcium and pERK signaling.[82] Similarly, biased responses involving pERK1/2 activation, plasma membrane ruffling, and IP_1 accumulation are seen for calcimimetics in the presence of PAMs acting on the calcium receptor.[122,123]

In light of the fact that NAMs and PAMs change the quality as well as the quantity of signals, additional parameters are required to characterize the pharmacological activities of these molecules. Thus, the bias imposed on natural signaling by PAMs should be quantified in the same manner as the direct signaling bias of agonists; a scale similar to the transducer coefficient scale [$\Delta\Delta\log(\tau/K_A)$ values] can be derived from the functional allosteric model to yield an index to do this, namely, $\Delta\log(\alpha\beta)$.

The response to an agonist A for a given signaling pathway (designated pathway 1) is given by the Black/Leff operational model;[8] see eq 10. Defining the response for one signaling pathway, the transduction coefficient[118] for agonism is given as

$$\log(\tau_{A1}/K_{A1}) \tag{58}$$

and in the presence of a saturating concentration of the positive allosteric modulator B, the response to the agonist is given as

$$\text{response}' = ([A]\beta_1\tau_{A1})/\{[A](1 + \beta_1\tau_{A1}) + K_{A1}/\alpha_1\} \tag{59}$$

In the presence of the allosteric modulator, the efficacy of agonist A is $\beta\tau_{A1}$, and the affinity of agonist A for the receptor is αK_{A1}. Under these circumstances, the transduction coefficient of the agonist in the presence of the allosteric modulator is $\log[\beta_1\tau_{A1}/(K_{A1}/\alpha_1)]$. Thus, the ratio of the logarithms of the transduction coefficients for agonism in the absence and presence of the modulator is given from eq 58 as

$$\Delta\log(\tau_{A1}/K_{A1})^{\text{modulator}} = \log(\alpha_1\beta_1) \tag{60}$$

Equations analogous to eqs 58−60 can be derived for the agonist as it activates a second signaling pathway (designated pathway 2), making the logarithm of the induced bias the difference in the $\log(\alpha\beta)$ values

$$\log[(\text{induced})\ \text{bias}] = \log(\alpha_1\beta_1) - \log(\alpha_2\beta_2) = \Delta\log(\alpha\beta) \tag{61}$$

and

$$\text{induced bias} = 10^{\Delta\log(\alpha\beta)} \tag{62}$$

This equation can be used to quantify the induced bias on natural signaling produced by any allosteric ligand through $\Delta\log(\alpha\beta)$. This index quantifies the relative allosteric effects of the agonist as it activates different signaling pathways. For example, the calcium receptor sensing PAM cinacalcet potentiates calcium-induced intracellular calcium release with an $\alpha\beta$ value of 3.0.[122] However, the $\alpha\beta$ potentiation for calcium-induced pERK1/2 effects is $\alpha\beta = 1.9$; this yields a small biased PAM effect toward intracellular calcium release of $10^{\Delta\log(\alpha\beta)} = 10^{0.2} = 1.58$ over pERK1/2 responses. Similarly, the mGlutamic 5 acid receptor NAM M-5MPEP blocks calcium oscillations with an $\alpha\beta$ value of 0.01 and inositol phosphate accumulation with an $\alpha\beta$ value of 0.37.[124] This indicates a strong biased antagonism of calcium oscillations of $10^{\Delta\log(\alpha\beta)} = 10^{1.57} = 37$, indicating that 5MPEP has 37 times the blocking activity for calcium oscillations as for inositol accumulation.

In general, consideration of the allosteric nature of 7TMRs has caused receptor models to evolve into systems that define interactions between cellular protein species and extracellular ligands. Although these synoptic systems are more complete and more correct physiologically, they are also more complex and require an added measure of experimentation and documentation to quantify and characterize drug activity; that is, a single drug might have different activities depending on the

Table 1. Biased Signaling for Dopamine D_{2L} Receptor Agonists

	cyclic AMP accumulation			stimulation of pERK1/2	
agonist	$\log(\tau/K_A)$	$\Delta\log(\tau/K_A)$	$\Delta\Delta\log(\tau/K_A)$	$\Delta\log(\tau/K_A)$	$\log(\tau/K_A)$
quinpirole	8.68				8.41
FAUC321	7.7	0.98	0.78[a]	0.22	8.19

[a]Bias = $10^{\Delta\Delta\log(\tau/K_A)} = 6.02$.

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other species with which the receptor interacts. Also germane to allosteric systems is the notion that there is a library of conformations that the receptor can adopt; standard linkage models that predefine the receptor species present become inadequate to describe multiconformational systems. In this regard, the introduction of molecular dynamics into receptor pharmacology has filled the void in the form of receptor probability models.

6. RECEPTOR PROBABILITY MODELS

As experimental evidence emerged to show that there are a number of receptor conformations that can take part in receptor signaling, linkage models (where the receptor species are identified and linked with energetically equivalent pathways) became increasingly inadequate for the description of pluridimensional efficacy. To fill this void, multistate models incorporating molecular dynamics can be used to estimate the probabilities of formation of different multiple receptors states. One simple model begins with one receptor state (referred to as $[R_o]$) and defines the affinities of a ligand A and a G-protein G for that state as[18,19]

$$^{A}k_o = [AR_o]/[R_o][A] \tag{63}$$

and

$$^{G}k_o = [GR_o]/[R_o][G] \tag{64}$$

The probability of the receptor being in that state is denoted as p_o, whereas the probability of the receptor forming another conformation R_1 is defined as p_1. The probability ratio for forming state R_1 versus R_o is given as j_1, where $j_1 = p_1/p_o$ and the value of j controls the energy of transition between the states. This defines the relative probability of forming state R_1 with ligand binding as $^{A}j_1 = {}^{A}p_1/{}^{A}p_o$ and with G-protein binding as $^{G}j_1 = {}^{G}p_1/{}^{G}p_o$. A vector b can be defined where b refers to the fractional stabilization of a state with binding of either ligand (defined $^{A}b_1 = {}^{A}j_1/j_i$) or G-protein ($^{G}b_1 = {}^{G}j_1/j_i$). The magnitude of b is characteristic for each receptor state; thus, the various values of b constitute ligand affinity and efficacy. With these probabilities and vectors, the following operators are defined

$$\Omega = 1 + \sum j_i \tag{65}$$

$$\Omega_A = 1 + \Omega \sum {}^{A}b_i p_i \tag{66}$$

$$\Omega_G = 1 + \Omega \sum {}^{G}b_i p_i \tag{67}$$

$$\Omega_{AG} = 1 + \Omega \sum {}^{A}b_i {}^{G}b_i p_i \tag{68}$$

where i refers to the specific conformational state and the superscripts G and A refer to the G-protein and ligand-bound forms, respectively. It can be shown that macroaffinity is given by

$$\text{macroaffinity (K)} = {}^{A}k_0 \Omega_A (\Omega)^{-1} \tag{69}$$

where the interaction free energy between the ligand and a reference microstate of the receptor is related to k_0 and a measure of efficacy is given by

$$\text{efficacy } (\alpha) = (\Omega\Omega_{AG})(\Omega_A\Omega_G)^{-1} \tag{70}$$

This model allows for the simulation of the effects of ligand binding on collections of conformations that might have differing pharmacological effects (ensembles). Figure 9 shows four natural ensembles of conformations that have physiological functions, specifically activation of $G_{\alpha i}$, $G_{\alpha s}$, and β-arrestin and receptor internalization. Also shown are bars representing ligand-stabilized conformations. Where a stabilized conformation coincides with a functionally active conformation, a pharmacological function is associated with that stabilization; this confers the pattern of efficacy on that ligand. Shown are two different ligands with different arrays of efficacy due to stabilization of different receptor conformations. These functions can be used to describe agonist functional selectivity where different agonists activate different portions of stimulus–response cascades through activation of the same receptor.[125]

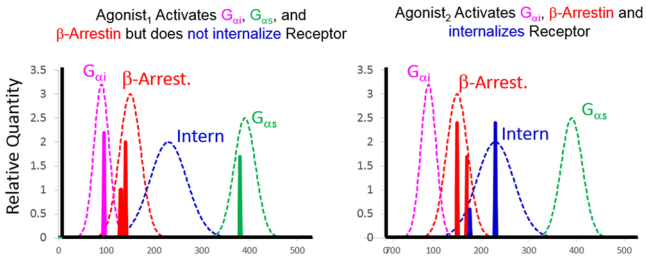

Figure 9. Ensemble view of a nonlinear scheme of efficacy for ligands. As ligands stabilize different conformations for which they have selective affinity (denoted as single bars), some of these conformations fall into natural ensembles that the receptor forms that have physiological functions (i.e., $G_{\alpha i}$, $G_{\alpha s}$, and β-arrestin activation and internalization). The functional conformations that are selectively stabilized by the ligand furnish the array of efficacies the ligand exhibits in pharmacological systems. Shown are two agonists with different arrays of efficacies.

7. NEW CHALLENGES: SYNOPTIC PHARMACOLOGY

As seen from the previous discussions, a great deal of research has gone into the description of receptor function and drug action; these efforts have been invaluable to the drug discovery process. However, relatively less effort has gone into the real-time kinetic description of receptor function, and because drug activity in vivo is very much a function of real time, this is a shortcoming of these "snapshot" models. Specifically, the interplay of different time scales for cellular responsiveness with changes in receptor conformation and transition might account for the still-unsolved capriciousness of host cell dependence on receptor function. In addition, the stoichiometric variance of receptor cofactors and cobinding bodies (i.e., signaling proteins) combined with biased ensembles of active-state receptors can produce very different relative potencies for agonists for the same receptor transfected into different host cells (i.e., calcitonin receptors in CHO and Cos cells,[126] dopamine receptors in U-77 and SK-N-MC cells[127]). To address these important effects, synoptic models must be employed that involve not only the receptor but also the receptor interactants in the cell. It will be difficult to attain a systematic and uniform system for classification of these effects, as a great deal of variation in coreactants with the receptor and a great number of possible agonist-stabilized biased receptor conformations are theoretically possible. A similar situation is seen with the classification of allosteric ligands where probe-dependent effects cause differences in receptor activity with both ligands and cell-signaling species.[128]

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8. SUMMARY AND CONCLUSIONS

Stephen Hawking's quotation at the beginning of this article cites equations as being the rules by which systems behave toward drugs and thus describe a "Universe" of interaction far beyond what can be explored experimentally. This can be enormously valuable, as predictions can then be made for all types of systems to explore therapeutic utility. Progress is made when experiments disprove and thus negate hypothesis-based models; this forces redefinition of the model to, presumably, a better model. Early representations of receptors (from the time of A.J. Clarke and others) modeled receptors as simple switches. Because of the fact that the experimental systems to test this model at the time were severely limited, this model sufficed for a time. Major advances in the description of 7TMR system were made through experiments showing that receptors floated in the cell membrane to associate with other proteins; this led to variations of the ternary complex model and furnished new parameters to characterize drug interactions with receptors. In addition, the operational model furnished a more physiological model with which to describe functional interactions of ligands with receptors. Elements of "two-state" theory from ion channels were introduced to accommodate the observation that receptors can demonstrate spontaneous activity; this finding also greatly expanded the concept of efficacy as a vector with negative as well as positive orientations. Further experimental refinements indicated that receptors are pleiotropically linked to often multiple signaling pathways. As assay technology progressed to furnish new ways to view receptor activity, the knowledge was developed that drugs can have multiple efficacies and that these efficacies need not be linearly linked to each other. This knowledge was evaluated in terms of data to show that receptors are allosteric entities that simply serve as conduits between ligands and cellular proteins. Quantitative models built to describe these allosteric interactions furnish parameters that are capable of quantifying drug activity on a system-to-system level and thus are more predictive than ever before in linking in vitro drug profiles with predicted therapeutic activity. As increasing numbers of allosteric molecules are being developed and tested in humans, it will be extremely interesting to see if the more quantitative models based on allostery will enable better drug candidate selection and a decrease in the failure rate of new drugs in the discovery and development process.

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Notes

The authors declare no competing financial interest.

Biography

Beginning his career as a synthetic chemist, Terry Kenakin received a Ph.D. in pharmacology at the University of Alberta, Edmonton, Canada. After a postdoctoral fellowship at University College London, U.K., he joined Burroughs-Wellcome as an Associate Scientist. From there, he continued working in drug discovery at Glaxo Inc., GlaxoWellcome, and finally GlaxoSmithKline. After leaving his position as a Director at GlaxoSmithKline Research and Development laboratories at Research Triangle Park, NC, Dr. Kenakin is now a professor in the Department of Pharmacology, University of North Carolina School of Medicine, Chapel Hill, NC. Currently, he is engaged in studies aimed at the optimal design of drug activity assay systems, the discovery and testing of allosteric molecules for therapeutic applications, and the quantitative modeling of drug effects. In addition, he is Director of the Pharmacology curriculum at the UNC School of Medicine. He is a member of numerous editorial boards, as well as Editor-in-Chief of *Journal of Receptors and Signal Transduction* and Co-Editor-in-Chief of *Current Opinion in Pharmacology*. He has authored numerous articles and has written 10 books on pharmacology.

REFERENCES

(1) Guldberg C. M.; Waage, P. Studies Concerning Affinity. *Forh.: Vidensk.-Selsk. Christiana* **1864**, 35.

(2) Hill, A. V. The Possible Effects of the Aggregation of the Molecules of Haemoglobin on its Dissociation Curves. *J. Physiol.* **1910**, *40* (Suppl), iv−vii.

(3) Langmuir, I. The Constitution and Fundamental Properties of Solids and Liquids. Part I. Solids. *J. Am. Chem. Soc.* **1916**, *38*, 2221−2295.

(4) Ariens, E. J. Affinity and Intrinsic Activity in the Theory of Competitive Inhibition. *Arch. Int. Pharmacodyn. Ther.* **1954**, *99*, 32−49.

(5) Ariens, E. J. *Molecular Pharmacology*; Academic Press: New York, 1964.

(6) Sastry, R.; Cheng, H. C. Dissociation Constants of D- and L-Lactoylcholines and Related Compounds at Cholinergic Receptors. *J. Pharmacol. Exp. Ther.* **1972**, *180*, 326−339.

(7) Stephenson, R. P. A Modification of Receptor Theory. *Br. J. Pharmacol. Chemother.* **1956**, *11*, 379−393.

(8) Black, J. W.; Leff, P. Operational Models of Pharmacological Agonism. *Proc. R. Soc. London, Ser. B* **1983**, *220*, 141−162.

(9) Michaelis, L.; Menten, M. L. Die Kinetik der Invertinwirkung. *Biochem. Z.* **1913**, *49*, 333−369.

(10) Johnson, K.; Goody, R. S. The Original Michaelis Constant: Translation of the 1913 Michaelis−Menten Paper. *Biochemistry* **2011**, *50* (39), 8264−8269.

(11) Black, J. W.; Leff, P.; Shankley, N. P.; Wood, J. An Operational Model of Pharmacological Agonism: The Effect of E/[A] Curve Shape on Agonist Dissociation Constant Estimation. *Br. J. Pharmacol.* **1985**, *84*, 561−571.

(12) Colquhoun, D. The Relationship between Classical and Cooperative Models for Drug Action. In *Symposium on Drug Receptors*; Rang, H. P., Ed.; University Park Press: Baltimore, MD, 1973; pp 149−182.

(13) Burgen, A. S. V. Conformational Changes and Drug Action. *Fed. Proc.* **1981**, *40*, 2723−2728.

(14) Fraunfelder, H.; Parak, F.; Young, R. D. Conformational Substrates in Proteins. *Annu. Rev. Biophys. Biophys. Chem.* **1988**, *17*, 451−79.

(15) Fraunfelder, H.; Sligar, S. G.; Wolynes, P. G. The Energy Landscapes and Motions of Proteins. *Science* **1991**, *254*, 1598−1603.

(16) Hilser, J.; Dowdy, D.; Oas, T. G.; Freire, E. The Structural Distribution of Cooperative Interactions in Proteins: Analysis of the Native State Ensemble. *Proc. Natl. Acad. Sci. U. S. A.* **1998**, *95*, 9903−9908.

(17) Hilser, V. J.; Garcia-Moreno, E. B.; Oas, T. G.; Kapp, G.; Whitten, S. T. A Statistical Thermodynamic Model of Protein Ensembles. *Chem. Rev.* **2006**, *106*, 1545−1558.

(18) Onaran, H. O.; Costa, T. Agonist Efficacy and Allosteric Models of Receptor Action. *Ann. N. Y. Acad. Sci.* **1997**, *812*, 98−115.

(19) Onaran, H. O.; Scheer, A.; Cotecchia, S.; Costa, T. A Look at Receptor Efficacy. From the Signaling Network of the Cell to the Intramolecular Motion of the Receptor. In *The Pharmacology of Functional, Biochemical, and Recombinant Systems*; Kenakin, T. P., Angus, J. A., Eds.; Springer: Berlin, 2002; Vol. *148*, Chapter 8, 217−280.

(20) Kenakin, T.; Miller, L. J. Seven Transmembrane Receptors as Shapeshifting Proteins: The Impact of Allosteric Modulation and

DOI: 10.1021/acs.chemrev.5b00561
Chem. Rev. 2017, 117, 4−20

Functional Selectivity on New Drug Discovery. *Pharmacol Rev.* **2010**, *62* (2), 265−304.
(21) Stockton, J. M.; Birdsall, N. J. M.; Burgen, A. S. V.; Hulme, E. C. Modification of the Binding Properties of Muscarinic Receptors by Gallamine. *Mol. Pharmacol.* **1983**, *23*, 551−557.
(22) Ehlert, E. J. Estimation of the Affinities of Allosteric Ligands Using Radioligand Binding and Pharmacological Null Methods. *Mol. Pharmacol.* **1988**, *33*, 187−194.
(23) Rasmussen, S. G. F.; DeVree, B. T.; Zou, Y.; Kruse, A. C.; Chung, K. Y.; Kobilka, T. S.; Thian, F. S.; Chae, P. S.; Pardon, E.; Calinski, D.; et al. Crystal Structure of the β2 Adrenergic Receptor−Gs Protein Complex. *Nature* **2011**, *477*, 549−555.
(24) Rasmussen, S. G. F.; Fung, J. J.; Choi, H.-J.; Pardon, E.; Casarosa, P.; Chae, P. S.; DeVree, B. T.; Rosenbaum, D. M.; Thian, F. S.; Kobilka, T. S.; et al. Structure of a Nanobody-Stabilized Active State of the β2 Adrenoceptor. *Nature* **2011**, *469*, 175−180.
(25) Mary, S.; Damian, M.; Louet, M.; Floquet, N.; Fehrentz, J.-A.; Marie, J.; Martinez, J.; Banères, J.-L. Ligands and Signaling Proteins Govern the Conformational Landscape explored by G Protein-Coupled Receptors. *Proc. Natl. Acad. Sci. U. S. A.* **2012**, *109*, 8304−8309.
(26) Yan, F.; Mosier, P. D.; Westkaemper, R. B.; Roth, B. L. Gα-subunits Differentially Alter the Conformation and Agonist Affinity of κ-Opioid Receptors. *Biochemistry* **2008**, *47*, 1567−1578.
(27) Black, J. W.; Shankley, N. P. Interpretation of Agonist Affinity Estimations: the Question of Distributed Receptor States. *Proc. R. Soc. London, Ser. B* **1990**, *240*, 503−518.
(28) Cuatrecasas, P. Membrane Receptors. *Annu. Rev. Biochem.* **1974**, *43*, 169−214.
(29) Hulme, E. C.; Birdsall, N. J. M.; Burgen, A. S. V.; Metha, P. The Binding of Antagonists to Brain Muscarinic Receptors. *Mol. Pharmacol.* **1978**, *14*, 737−750.
(30) Lefkowitz, R. J.; Mullikin, D.; Caron, M. G. Regulation of β-Adrenergic Receptors by Guanyl-5′-yl imidodiphosphate and Other Purine Nucleotides. *J. Biol. Chem.* **1976**, *251*, 4686−4692.
(31) Maguire, M. E.; Van Arsdale, P. M.; Gilman, A. G. An Agonist-Specific Effect of Guanine Nucleotides on the Binding of the Beta Adrenergic Receptor. *Mol. Pharmacol.* **1976**, *12*, 335−339.
(32) DeLean, A.; Stadel, J. M.; Lefkowitz, R. J. A Ternary Complex Model Explains the Agonist Specific Binding Properties of Adenylate Cyclase Coupled β-Adrenergic Receptor. *J. Biol. Chem.* **1980**, *255*, 7108−7117.
(33) Costa, T.; Herz, A. Antagonists with Negative Intrinsic Activity at δ-Opioid Receptors Coupled to GTP-Binding Proteins. *Proc. Natl. Acad. Sci. U. S. A.* **1989**, *86*, 7321−7325.
(34) Samama, P.; Cotecchia, S.; Costa, T.; Lefkowitz, R. J. A Mutation-Induced Activated State of the β2-Adrenergic Receptor: Extending the Ternary Complex Model. *J. Biol. Chem.* **1993**, *268*, 4625−4636.
(35) Weiss, J. M.; Morgan, P. H.; Lutz, M. W.; Kenakin, T. P. The Cubic Ternary Complex Receptor−Occupancy Model. I. Model Description. *J. Theor. Biol.* **1996**, *178*, 151−167.
(36) Weiss, J. M.; Morgan, P. H.; Lutz, M. W.; Kenakin, T. P. The Cubic Ternary Complex Receptor−Occupancy Model. II. Understanding Apparent Affinity. *J. Theor. Biol.* **1996**, *178*, 169−182.
(37) Weiss, J. M.; Morgan, P. H.; Lutz, M. W.; Kenakin, T. P. The Cubic Ternary Complex Receptor-Occupancy Model. III. Resurrecting Efficacy. *J. Theor. Biol.* **1996**, *181*, 381−397.
(38) Bouaboula, M.; Perrachon, S.; Milligan, L.; Canat, X.; Rinaldi-Carmona, M.; Portier, M.; Barth, F.; Calandra, B.; Pecceu, F.; Lupker, J.; et al. A Selective Inverse Agonist for Central Cannabinoid Receptor Inhibits Mitogen-Activated Protein Kinase Activation Stimulated by Insulin or Insulin-Like Growth Factor. *J. Biol. Chem.* **1997**, *272*, 22330−22339.
(39) Langley, J. N. On the Physiology of Salivary Secretion. *J. Physiol. (Oxford, U. K.)* **1878**, *1*, 339−369.
(40) Clark, A. J. *The Mode of Action of Drugs on Cells*; Edward Arnold: London, 1933.
(41) Gaddum, J. H. Theories of Drug Antagonism. *Pharmacol. Rev.* **1957**, *9*, 211−218.
(42) Del Castillo, J.; Katz, B. Interaction at End-Plate Receptors between Different Choline Derivatives. *Proc. R. Soc. London, Ser. B* **1957**, *146*, 369−381.
(43) Katz, B.; Thesleff, S. A study of the "Desensitization" Produced by Acetylcholine at the Motor End-plate. *J. Physiol. (Oxford, U. K.)* **1957**, *138*, 63−80.
(44) Changeux, J.-P.; Thiery, J.; Tung, Y.; Kittel, C. On the Cooperativity of Biological Membranes. *Proc. Natl. Acad. Sci. U. S. A.* **1967**, *57*, 335−341.
(45) Karlin, A. On the Application of a "Plausible Model" of Allosteric Proteins to the Receptor for Acetylcholine. *J. Theor. Biol.* **1967**, *16*, 306−320.
(46) Thron, C. D. On the Analysis of Pharmacological Experiments in Terms of an Allosteric Receptor Model. *Mol. Pharmacol.* **1973**, *9*, 1−9.
(47) MacKay, D. Use of Null Equations, Based on Classical Receptor and Ternary Models of Drug Action, to Classify Receptors and Receptor−Effector Systems. In *Perspectives on Receptor Classification*; Black, J. W., Jenkinson, D. H., Gerskowitch, V. P., Eds.; Alan R. Liss: New York, 1987; pp 193−206.
(48) Mackay, D. Interpretation of Relative Potencies, Relative Efficacies and Apparent Affinity Constants of Agonist Drugs Estimated from Concentration−Response Curves. *J. Theor. Biol.* **1990**, *142* (3), 415−27.
(49) Mayo, K. H.; Nunez, M.; Burke, C.; Starbuck, C.; Lauffenberger, D.; Savage, C. R., Jr. Epidermal Growth Factor Receptor Binding is Not a Simple One-Step Process. *J. Biol. Chem.* **1989**, *264*, 17838−17844.
(50) Ross, E. M. Signal Sorting and Amplification Through G Protein-Coupled Receptors. *Neuron* **1989**, *3*, 141−152.
(51) Birnbaumer, L. G.; Yatani, A.; VanDongen, A. M. J.; Graf, R.; Codina, J.; Odabe, K.; Mattera, R.; Brown, A. M. G Protein Coupling of Receptors to Ionic Channels and Other Effector Systems. In *G-Proteins and Signal Transduction*; Nathanson, N. M., Harden, T. K., Eds.; Rockefeller University Press: New York, 1990; pp 169−183.
(52) Bourne, H. R.; Sanders, D. A.; McCormick, F. The GTPase Superfamily: A Conserved Switch for Diverse Cell Functions. *Nature* **1990**, *348*, 125−132.
(53) Podleski, T. R.; Changeux, J.-P. On the Excitability and Cooperativity of the Electroplax Membrane. In *Fundamental Concepts in Drug−Receptor Interaction*; Danielli, J. F., Moran, J. F., Triggle, D. J., Eds.; Academic Press: New York, 1970; pp 93−119.
(54) Ross, E. M.; Maguire, M. E.; Sturgill, T. W.; Biltonen, R. L.; Gilman, A. G. Relationship between the β-Adrenergic Receptor and Adenylate Cyclase. *J. Biol. Chem.* **1977**, *252*, 5761−5775.
(55) Heidenreich, K. A.; Weiland, G. A.; Molinoff, P. B. Characterization of Radiolabeled Agonist Binding to β-Adrenergic Receptors in Mammalian Tissues. *J. Cyclic Nucleotide Res.* **1980**, *6*, 217−230.
(56) Iyengar, R.; Abramowitz, J.; Bordelon-Riser, M.; Birnbaumer, L. Hormone Receptor Mediated Stimulation of Adenylyl Cyclase Systems. Nucleotide Effects and Analysis in Terms of a Simple Two-State Model for the Basic Receptor-Affected Enzyme. *J. Biol. Chem.* **1980**, *255*, 3558−3564.
(57) Jacobs, S.; Cuatrecasas, P. The Mobile Receptor Hypothesis and "Cooperativity" of Hormone Binding: Applications to Insulin. *Biochim. Biophys. Acta, Biomembr.* **1976**, *433*, 482−495.
(58) Boeynaems, J. M.; Dumont, J. E. The Twostep Model of Ligand-Receptor Interaction. *Mol. Cell. Endocrinol.* **1977**, *7*, 33−47.
(59) Wregget, K. A.; De Lean, A. The Ternary Complex Model: Its Properties and Application to Ligand Interactions with the D2-Dopamine Receptor and the Anterior Pituitary Gland. *Mol. Pharmacol.* **1984**, *26*, 214−227.
(60) Ehlert, F. J. The Relationship between Muscarinic Receptor Occupancy and Adenylate Cyclase Inhibition in the Rabbit Myocardium. *Mol. Pharmacol.* **1985**, *28*, 410−421.

DOI: 10.1021/acs.chemrev.5b00561
Chem. Rev. 2017, 117, 4−20

(61) Neubig, R. R.; Gantzos, R. D.; Thomsen, W. J. Mechanism of Agonist and Antagonist Binding to α2-Andrenergic Receptors: Evidence for a Precoupled Receptor Guanine Nucleotide Protein Complex. *Biochemistry* **1988**, *27*, 2374−2384.
(62) Minton, A. P.; Sokolovsky, M. A Model for the Interaction of Muscarinic Receptors, Agonists, and Two Distinct Effector Substances. *Biochemistry* **1990**, *29*, 1586−1593.
(63) Costa, T.; Ogino, Y.; Munson, P. J.; Onaran, H. O.; Rodbard, D. Drug Efficacy at Guanine Nucleotide-Binding Regulatory Protein-Linked Receptors: Thermodynamic Interpretation of Negative Antagonism and of Receptor Activity in the Absence of Ligand. *Mol. Pharmacol.* **1992**, *41*, 549−560.
(64) Lefkowitz, R. J.; Cotecchia, S.; Samama, P.; Costa, T. Constitutive Activity of Receptors Coupled to Guanine Nucleotide Regulatory Proteins. *Trends Pharmacol. Sci.* **1993**, *14*, 303−307.
(65) Koshland, D. E. The Active Site of Enzyme Action. *Adv. Enzymol* **2006**, *22*, 45−97.
(66) Umbarger, H. E. Evidence for a Negative Feedback Mechanism in the Biosynthesis of Isoleucine. *Science* **1956**, *123*, 848.
(67) Lee, N. H.; Hu, J.; El-Fakahany, E. E. Modulation by Certain Conserved Aspartate Residues of the Allosteric Interaction of Gallamine and the Ml Muscarinic Receptor. *J. Pharmacol. Exp. Ther.* **1992**, *262*, 312−316.
(68) Oikonomakos, N. G.; Skamnaki, V. T.; Tsitsanou, K. E.; Gavalas, N. G.; Johnson, L. A New Allosteric Site in Glycogen Phosphorylase B as a Target for Drug Interactions. *Structure* **2000**, *8*, 575−584.
(69) Pargellis, C.; Tong, L.; Churchill, L.; Cirillo, P. F.; Gilmore, T.; Graham, A. G.; Grob, P. M.; Hickey, E. R.; Moss, N.; Pav, S.; Regan, J. Inhibition of p38 MAP Kinase by Utilizing a Novel Allosteric Binding Site. *Nat. Struct. Biol.* **2002**, *9*, 268−272.
(70) Kenakin, T. P. New concepts in drug discovery: Collateral Efficacy and Permissive Antagonism. *Nat. Rev. Drug Discovery* **2005**, *4*, 919−927.
(71) Ehlert, F. J. Analysis of Allosterism in Functional Assays. *J. Pharmacol. Exp. Ther.* **2005**, *315*, 740−754.
(72) Price, M. R.; Baillie, G. L.; Thomas, A.; Stevenson, L. A.; Easson, M.; Goodwin, R.; McLean, A.; Mclntosh, L.; Goodwin, G.; Walker, G.; et al. Allosteric Modulation of the Cannabinoid CB1 Receptor. *Mol. Pharmacol.* **2005**, *68*, 1484−1495.
(73) Maeda, K.; Nakata, H.; Koh, Y.; Miyakawa, T.; Ogata, H.; Takaoka, Y.; Shibayama, S.; Sagawa, K.; Fukushima, D.; Moravek, J.; et al. Spirodiketopiperazine-Based CCR5 Inhibitor Which Preserves CC-Chemokine/CCR5 Interactions and Exerts Potent Activity Against R5 Human Immunodeficiency Virus Type 1 in Vitro. *J. Virol.* **2004**, *78*, 8654−8662.
(74) Watson, C.; Jenkinson, S.; Kazmierski, W.; Kenakin, T. P. The CCR5 Receptor-Based Mechanism of Action of 873140, a Potent Allosteric Non-Competitive HIV Entry-Inhibitor. *Mol. Pharmacol.* **2005**, *67*, 1268−1282.
(75) Litschig, S.; Gasparini, T.; Ruegg, D.; Stoehr, N.; Flor, P. J.; Vranesic, I.; Prézeau, L.; Pin, J. P.; Thomsen, C.; Kuhn, R. CPCCOEt, a Noncompetitive Metabotropic Glutamate Receptor 1 Antagonist, Inhibits Receptor Signaling without Affecting Glutamate Binding. *Mol. Pharmacol.* **1999**, *55*, 453−461.
(76) Trankle, C.; Weyand, A.; Schroter, A.; Mohr, K. Using a Radioalloster to Test Predictions of the Cooperativity Model for Gallamine Binding to the Allosteric site of Muscarinic Acetylcholine (M2) Receptors. *Mol. Pharmacol.* **1999**, *56*, 962−965.
(77) Kew, J. N.; Trube, G.; Kemp, J. A. A Novel Mechanism of Activity-Dependent NMDA Receptor Antagonism Describes the Effect of Ifenprodil inR Cultured Cortical Neurons. *J. Physiol.* **1996**, *497*, 761−772.
(78) Hudson, B. D.; Christiansen, E.; Murdoch, H.; Jenkins, L.; Hansen, A. H.; Madsen, O.; Ulven, T.; Milligan, G. Complex Pharmacology of Novel Allosteric Free Fatty Acid 3 Receptor Ligands. *Mol. Pharmacol.* **2014**, *86* (2), 200−210.
(79) Maelicke, A.; Albuquerque, E. X. New Approach to Drug Therapy of Alzheimer's Dementia. *Drug Discovery Today* **1996**, *1*, 53−59.
(80) Krause, R. M.; Buisson, B.; Bertrand, S.; Corringer, P.-J.; Galzi, J.-L.; Changeux, J.-P.; Bertrand, D. Ivermectin: A Positive Allosteric Effector of the α7 Neuronal Nicotinic Acetylcholine Receptor. *Mol. Pharmacol.* **1998**, *53*, 283−294.
(81) Suratman, S.; Leach, K.; Sexton, P. M.; Felder, C. C.; Loiacono, R. E.; Christopoulos, A. Impact of Species Variability and 'Probe-Dependence' on the Detection and in vivo Validation of Allosteric Modulation at the M4Muscarinic Acetylcholine Receptor. *Br. J. Pharmacol.* **2011**, *162*, 1659−1670.
(82) Koole, C.; Wootten, D.; Simms, J.; Valant, C.; Sridhar, R.; Woodman, O. L.; Miller, L. J.; Summers, R. J.; Christopoulos, A.; Sexton, P. M. Allosteric Ligands of the Glucagon-like Peptide 1 Receptor (GLP-1R) Differentially Modulate Endogenous and Exogenous Peptide Responses in a Pathway-Selective Manner: Implications for Drug Screening. *Mol. Pharmacol.* **2010**, *78*, 456−465.
(83) Gonzalez, E.; Kulkarni, H.; Bolivar, H.; Mangano, A.; Sanchez, R. The Influence of CCL3L1 Gene-Containing Segmental Duplications on HIV-1/AIDS Susceptibility. *Science* **2005**, *307*, 1434−1440.
(84) Muniz-Medina, V. M.; Jones, S.; Maglich, J. M.; Galardi, C.; Hollingsworth, R. E.; Kazmierski, W. M.; Ferris, R. G.; Edelstein, M. P.; Chiswell, K. E.; Kenakin, T. P. The Relative Activity of 'Function Sparing' HIV-1 Entry Inhibitors on Viral Entry and CCR5 Internalization: Is Allosteric Functional Selectivity a Valuable Therapeutic Property? *Mol. Pharmacol.* **2009**, *75*, 490−501.
(85) Trkola, A.; Kuhmann, S. E.; Strizki, J. M.; Maxwell, E.; Ketas, T.; Morgan, T.; Pugach, P.; Xu, S.; Wojcik, L.; Tagat, J.; et al. HIV-1 Escape from a Small Molecule, CCR5-Specific Entry Inhibitor Does Not Involve CXCR4 Use. *Proc. Natl. Acad. Sci. U. S. A.* **2002**, *99*, 395−400.
(86) Kuhmann, S. E.; Pugach, P.; Kunstman, K. J.; Taylor, J.; Stanfield, R. L.; Snyder, A.; Strizki, J. M.; Riley, J.; Baroudy, B. M.; Wilson, I. A.; et al. Genetic and Phenotypic Analyses of Human Immunodeficiency Virus Type 1 Escape from Small-Molecule CCR Inhibitor. *J. Virol.* **2004**, *78*, 2790−2807.
(87) Kenakin, T. P. Collateral Efficacy in Drug Discovery: Taking Advantage of the Good (Allosteric) Nature of 7TM Receptors. *Trends Pharmacol. Sci.* **2007**, *28*, 407−415.
(88) Ellis, J.; Huyler, J.; Brann, M. R. Allosteric Regulation of Cloned M1-M5Muscarinic Receptor Subtypes. *Biochem. Pharmacol.* **1991**, *42*, 1927−1932.
(89) Melchiorre, C.; Minarini, A.; Angeli, P.; Giardina, D.; Gulini, U.; Quaglia, W. Polymethylene Tetraamines as Muscarinic Receptor Probes. *Trends Pharmacol. Sci.* **1989**, *10* (Suppl), 55−59.
(90) Liang, J. S.; Carsi-Gabrenas, J.; Krajewski, J. L.; McCafferty, J. M.; Purkerson, S. L.; Santiago, M. P.; Strauss, W. L.; Valentine, H. H.; Potter, L. T. Anti-Muscarinic Toxins from *Dendroaspis angusticeps*. *Toxicon* **1996**, *34*, 1257−1267.
(91) Gnagey, A. L.; Seidenberg, M.; Ellis, J. Site-Directed Mutagenesis Reveals Two Epitopes Involved in the Subtype Selectivity of the Allosteric Interactions of Gallamine at Muscarinic Acetylcholine Receptors. *Mol. Pharmacol.* **1999**, *56*, 1245−1253.
(92) Johnson, M. P.; Nisenbaum, E. S.; Large, T. H.; Emkey, R.; Baez, M.; Kingston, A. E. Allosteric Modulators of Metabotropic Glutamate Receptors: Lessons Learned from mGlu1, mGlu2, and mGlu5 Potentiators and Agonists. *Biochem. Soc. Trans.* **2004**, *32*, 881−888.
(93) Bouvier, M. Oligomerization of G-Protein Coupled Transmitter Receptors. *Nature Rev. Neurosci.* **2001**, *2*, 274−286.
(94) James, J. R.; Oliveira, M. I.; Carmo, A. M.; Iaboni, A.; Davis, S. J. A Rigorous Experimental Framework for Detecting Protein Oligomerization Using Bioluminescence Resonance Energy Transfer. *Nat. Methods* **2006**, *3*, 1001−1006.
(95) Milligan, G.; Canals, M.; Pediani, J. D.; Ellis, J.; Lopez-Gimenez, J. F. The Role of GPCR Dimerisation/Oligomerisation in Receptor Signalling. *Ernst Schering Res. Found. Workshop* **2006**, *2006/2*, 145−162.

DOI: 10.1021/acs.chemrev.5b00561
Chem. Rev. 2017, 117, 4−20

(96) Hay, D. L.; Poyner, D. R.; Sexton, P. M. GPCR Modulation by RAMPS. *Pharmacol. Ther.* **2006**, *109*, 173−197.
(97) Franco, R.; Casadó, V.; Cortés, A.; Mallol, J.; Ciruela, F.; Ferré, S.; Lluis, C.; Canela, E. I. G-protein-coupled Receptor Heteromers: Function and Ligand Pharmacology. *Br. J. Pharmacol.* **2008**, *153*, S90−S98.
(98) Milligan, G. A day in the Life of a G Protein Coupled Receptor: The Contribution to Function of G Protein-coupled Receptor Dimerization. *Br. J. Pharmacol.* **2008**, *153*, S216−S229.
(99) Gurevich, V. V.; Gurevich, E. V. How and Why Do GPCRs Dimerize? *Trends Pharmacol. Sci.* **2008**, *29*, 234−240.
(100) Angers, S.; Salahpour, A.; Bouvier, M. Dimerization: An Emerging Concept for G Protein-Coupled Receptor Ontogeny and Function. *Annu. Rev. Pharmacol. Toxicol.* **2002**, *42*, 409−435.
(101) Giraldo, J. On the Fitting of Binding Data When Receptor Dimerization is Suspected. *Br. J. Pharmacol.* **2008**, *155*, 17−23.
(102) Kenakin, T. P. Agonist−Receptor Efficacy II: Agonist Trafficking of Receptor Signals. *Trends Pharmacol. Sci.* **1995**, *16*, 232−238.
(103) Kenakin, T. P. Perspectives in Pharmacology: Functional Selectivity and Biased Receptor Signaling. *J. Pharmacol. Exp. Ther.* **2011**, *336*, 296−302.
(104) Kenakin, T. P. The Potential for Selective Pharmacological Therapies Through Biased Receptor Signaling. *BMC Pharmacol. Toxicol.* **2012**, *13*, 3−8.
(105) Kenakin, T.; Christopoulos, A. Signalling Bias in New Drug Discovery: Detection, Quantification and Therapeutic Impact. *Nat. Rev. Drug Discovery* **2013**, *12* (3), 205−216.
(106) Mailman, R. B.; Lawler, C. P.; Lewis, M. M.; Blake, B. Functional Effects of Novel Dopamine Ligands: Dihydrexidine and Parkinson's Disease as a First Step. In *Dopamine Receptor Subtypes: From Basic Science to Clinical Application*; Jenner, P., Demirdamar, R., Eds.; IOS Press: Fairfax, VA, 1998.
(107) Berg, K. A.; Maayani, S.; Goldfarb, J.; Scaramellini, C.; Leff, P.; Clarke, W. P. Effector Pathway-Dependent Relative Efficacy at Serotonin Type 2A and 2C Receptors: Evidence for Agonist-Directed Trafficking of Receptor Stimulus. *Mol. Pharmacol.* **1998**, *54*, 94−104.
(108) Violin, J. D.; Dewire, S. M.; Barnes, W. G.; Lefkowitz, R. J. G Protein-Coupled Receptor Kinase and Beta-Arrestin-Mediated Desensitization of the Angiotensin II Type 1A Receptor Elucidated by Diacylglycerol Dynamics. *J. Biol. Chem.* **2006**, *281*, 36411−36419.
(109) Violin, J. D.; DeWire, S. M.; Yamashita, D.; Rominger, D. H.; Nguyen, L.; Schiller, K.; Whalen, E. J.; Gowen, M.; Lark, M. W. Selectively Engaging β-Arrestins at the Angiotensin II Type 1 Receptor Reduces Blood Pressure and Increases Cardiac Performance. *J. Pharmacol. Exp. Ther.* **2010**, *335*, 572−579.
(110) White, K. L.; Scopton, A. P.; Rives, M.-L.; Bikbulatov, R. V.; Polepally, P. R.; Brown, P. J.; Kenakin, T.; Javitch, J. A.; Zjawiony, J. K.; Roth, B. L. Identification of Novel Functionally Selective k-Opioid Receptor Scaffolds. *Mol. Pharmacol.* **2014**, *85*, 83−90.
(111) Raehal, K. M.; Walker, J. K. L.; Bohn, L. M. Morphine Side Effects in β-Arrestin 2 Knockout Mice. *J. Pharmacol. Exp. Ther.* **2005**, *314*, 1195−1201.
(112) DeWire, S. M.; Yamashita, D. S.; Rominger, D. H.; Liu, G.; Cowan, C. L.; Graczyk, T. M.; Chen, X.-T.; Pitis, P. M.; Gotchev, D.; Yuan, C.; et al. A G Protein-Biased Ligand at the μ-Opioid Receptor Is Potently Analgesic with Reduced Gastrointestinal and Respiratory Dysfunction Compared with Morphine. *J. Pharmacol. Exp. Ther.* **2013**, *344*, 708−717.
(113) Charfi, I.; Audet, N.; Bagheri Tudashki, H.; Pineyro, G. Identifying Ligand-Specific Signalling Within Biased Responses: Focus on δ Opioid Receptor Ligands. *Br. Pharmacol.* **2015**, *172*, 435−448.
(114) Ferrari, S. L.; Pierroz, D. D.; Glatt, V.; Goddard, D. S.; Bianchi, E. N.; Lin, F. T.; Manen, D.; Bouxsein, M. L. Bone Response to Intermittent Parathyroid Hormone is Altered in Mice Null for β-Arrestin2. *Endocrinology* **2005**, *146*, 1854−1862.
(115) Gesty-Palmer, D.; Chen, M.; Reiter, E.; Ahn, S.; Nelson, C. D.; Wang, S.; Eckhardt, A. E.; Cowan, C. L.; Spurney, R. F.; Luttrell, L. M.; et al. Distinct -Arrestin- and G Protein Dependent Pathways for Parathyroid Hormone Receptor-Stimulated ERK1/2 Activation. *J. Biol. Chem.* **2006**, *281*, 10856−10864.
(116) Gesty-Palmer, D.; Flannery, P.; Yuan, L.; Corsino, L.; Spurney, R.; Lefkowitz, R. J.; Luttrell, L. M. A β-Arrestin-Biased Agonist of the Parathyroid Hormone Receptor (PTH1R) Promotes Bone Formation Independent of G Protein Activation. *Sci. Transl. Med.* **2009**, *1*, 1ra1.
(117) Sonoda, N.; Imamura, T.; Yoshizaki, T.; Babendure, J. L.; Lu, J.-C.; Olefsky, J. M. β-Arrestin-1 Mediates Glucagon-like Peptide-1 Signaling to Insulin Secretion in Cultured Pancreatic β Cells. *Proc. Natl. Acad. Sci. U. S. A.* **2008**, *105*, 6614−6619.
(118) Kenakin, T.; Watson, C.; Muniz-Medina, V.; Christopoulos, A.; Novick, S. A Simple Method for Quantifying Functional Selectivity and Agonist Bias. *ACS Chem. Neurosci.* **2012**, *3*, 193−203.
(119) Tschammer, N.; Bollinger, S.; Kenakin, T.; Gmeiner, P. Histidine 6.55 is a Major Determinant of Ligand-biased Signaling in Dopamine D2L Receptor. *Mol. Pharmacol.* **2011**, *79* (3), 575−85.
(120) Maillet, E. L.; Pellegrini, N.; Valant, C.; Bucher, B.; Hibert, M.; Bourguignon, J. J.; Galzi, J. L. A Novel, Conformation-specific Allosteric Inhibitor of the Tachykinin NK2 Receptor (NK2R) with Functionally Selective Properties. *FASEB J.* **2007**, *21*, 2124−2134.
(121) Mathiesen, J. M.; Ulven, T.; Martini, L.; Gerlach, L. O.; Heinemann, A.; Kostenis, E. Identification of Indole Derivatives Exclusively Interfering with a G Protein-Independent Signaling Pathway of the Prostaglandin D2 Receptor CRTH2. *Mol. Pharmacol.* **2005**, *68*, 393−402.
(122) Davey, A. E.; Leach, K.; Valant, C.; Conigrave, A. D.; Sexton, P. M.; Christopoulos, A. Positive and Negative Allosteric Modulators Promote Biased Signaling at the Calcium-sensing Receptor. *Endocrinology* **2012**, *153*, 1232−1241.
(123) Cook, A. E.; Mistry, S. N.; Gregory, K. J.; Furness, S. G. B.; Sexton, P. M.; Scammells, P. J.; Conigrave, A. D.; Christopoulos, A.; Leach, K. Biased Allosteric Modulation at the CaS Receptor Engendered by Structurally Diverse Calcimimetics. *Br. J. Pharmacol.* **2015**, *172*, 185−200.
(124) Bradley, S. J.; Langmead, C. J.; Watson, J. M. *Mol. Pharmacol.* **2011**, *79*, 874−885.
(125) Kenakin, T.P. Collateral Efficacy as a Pharmacological Problem Applied to New Drug Discovery. *Expert Opin. Drug Discovery* **2006**, *1*, 635−652.
(126) Christmanson, L.; Westermark, P.; Betsholtz, C. Islet Amyloid Polypeptide Stimulates Cyclic AMP Accumulation Via the Porcine Calcitonin Receptor. *Biochem. Biophys. Res. Commun.* **1994**, *205*, 1226−1235.
(127) Peters, M. F.; Scott, C. W. Evaluating Cellular Impedance Assays for Detection of GPCR Pleiotropic Signaling and Functional Selectivity. *J. Biomol. Screening* **2009**, *14*, 246−255.
(128) Christopoulos, A.; Changeux, J.-P.; Catterall, W. A.; Fabbro, D.; Burris, T. P.; Cidlowski, J. A.; Olsen, R. W.; Peters, J. A.; Neubig, R. R.; Pin, J.-P.; Sexton, P. M.; Kenakin, T. P.; Ehlert, F. J.; Spedding, M.; Langmead, C. J. International Union of Basic and Clinical Pharmacology. XC. Multisite Pharmacology: Recommendations for the Nomenclature of Receptor Allosterism and Allosteric Ligands. *Pharmacol. Rev.* **2014**, *66*, 918−947.

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